



The Aninda Studio guidebook

অনিন্দ্য স্টুডিও

The mark, the colour, the type, the components, the words, the licences, and the honest list of what is missing. One file, no network.

Version 1.0.0 · sources verified 14 August 2026 · Aninda Sundar Howlader, Barishal, Bangladesh · aninda.sh15@gmail.com

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Contents

01 **Welcome** · স্বাগতম

What this is, who wrote it, and why the brand and the design system are two artefacts rather than one.

02 **The name** · নাম

Aninda Studio in two scripts, and the one difference between them that looks like a mistake and is not.

03 **The mark** · চিহ্ন

A circle tangent to a stem that overruns downward. One geometry, two weights, and the mistake that produced it.

04 **Icons** · আইকন

One rounded icon on every surface, Apple included — and exactly what that trades away.

05 **Colour** · রং

Every ramp, every role and every measured ratio in four themes, read out of the token files.

06 **Type** · হরফ

Two scripts, one scale, a measured multiplier for Bangla and a floor it never goes below.

07 **Space and shape** · ফাঁক ও আকার

A 4 px scale in ten steps, four radii, four target sizes and one focus ring.

08 **Components** · উপাদান

Thirty cards in three groups, and the three rules the component layer is built to keep.

09 **Motion** · গতি

Two durations, three curves, and why Material's spring system was read and not adopted.

10 **Voice** · কণ্ঠস্বর

Plain international English, British spelling, and the six words that are banned outright.

11 **Writing in Bangla** · বাংলায় লেখা

Standard Bangla by the Bangla Academy's own rules, and the policy of never inventing any.

12 **Applying it** · কাজে লাগানো

How to install it, theme it, check it, and rebuild this book.

13 **Licence and trademarks** · লাইসেন্স ও ট্রেডমার্ক

Three licences, one thing with no licence at all, and where the boundary falls.

14 **What this system does not do** · যা এই পদ্ধতি করে না

The honest list. Not a disclaimer — every item changes how much weight to put on something earlier.

– **Every file, in this file**

The whole kit, with a download link for each file.

Welcome • স্বাগতম

What this is, who wrote it, and why the brand and the design system are two artefacts rather than one.

English

This is the whole of Aninda Studio in one file. The mark, the colour, the type, the components, the words, the licences and the honest list of what is missing. Nothing here points at a server. If you have this file, you have the brand.

I am Aninda Sundar Howlader. I work alone, from Barishal, in Bangladesh. The studio serves two audiences in two scripts — Bangla and English — and most of the decisions in this book exist because those two scripts do not behave the same way.

Two halves, and why they are separate

Apple and Google both publish a design system openly, and both keep their brand book private. That is worth stating early, because it is the thing most often missed by anyone reading the Human Interface Guidelines and assuming it is a brand book. It is not. It is a platform design system. The brand rules — how the mark is drawn, when it may be recoloured, who may use it — are the part nobody published, and so they are the part that gets improvised.

This kit is deliberately two artefacts with two licences:

The two halves of the kit, and the licence each carries.

Part	What is in it	Licence
The design system	Tokens, colour roles, type scale, space, shape, motion, components	Apache-2.0
The identity	The name, the mark, the wordmark, the tile, the lockups	Not licensed at all
The writing	The chapters of this book	PolyForm Noncommercial 1.0.0

The design system has to be usable by someone who is not permitted to use the mark. If removing the mark broke the token set, the boundary would have been drawn in the wrong place. Chapter 13 sets out the terms in full.

How to read this

Each chapter has an English section and a Bangla one. The button at the top of the page shows one and hides the other. It is one book, not two, because two files drift apart and this whole project is arranged against drift.

Every number in the generated chapters — colour, type, space and shape, components, motion — was read out of the token files at build time. Not one of them was typed by hand into this book. If a token changes and this book is not rebuilt, the build's own check fails.

Where a technical term is the correct one, it stays, and it gets one plain sentence of explanation the first time it appears.

What was built, and in what order

Each stage of the work, and what it produced.

Stage	What it produced
Research	A benchmark against Apple, Google, WCAG 2.2, DTCG and the font licences
Strategy	The English writing standard
Directions	Four brand directions, compared as rendered pages rather than as descriptions
The mark	One geometry, two weights, ten files
Colour	Six ramps computed in OKLCH, then measured against every surface
Type	Thirty families measured, one pairing chosen on the measurements
Tokens	DTCG 2025.10, four themes, every colour carrying its own proof
Components	Thirty cards, each a self-contained HTML file
This book	The whole thing, assembled from those sources

Dates

The system is version 1.0.0. Every figure in it was measured or verified on 14 August 2026. Anything dated after that supersedes this file.

Bangla · বাংলা

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The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

The studio, in its own words

This sentence is the register benchmark for every Bangla string in the system. It was written in Bangla rather than translated, and reviewed against the Bangla Academy's 2012 rules and its dictionary.

আমি ছোটো, যত্নে গড়া সফটওয়্যার বানাই। কোনো কিছুর সীমা থাকলে সেটা এখানেই
লেখা থাকবে — লুকিয়ে রাখা হবে না।

In English: I build small, carefully made software. If something has a limit, it is written here — it will not be hidden.

The name • নাম

Aninda Studio in two scripts, and the one difference between them that looks like a mistake and is not.

English

The studio is called **Aninda Studio** in English and অনিন্দ্য স্টুডিও in Bangla.

aninda studio

Latin — aninda studio

অনিন্দ্য স্টুডিও

Bangla — অনিন্দ্য স্টুডিও

The one thing that looks like an inconsistency, and is not

The Bangla is অনিন্দ্য. The Latin is **aninda**. Those are not the same word letter for letter, and a careful reader will notice. Here is the answer in one line: অনিন্দ্য transliterates as *anindya*, and *aninda* is the ordinary everyday spelling the name has always been written with in Latin script. It is transliteration, not a mistake, and I would rather say so here than let someone find it and wonder.

Both forms are correct. Neither is a shortened version of the other. Use the Bangla in Bangla text and the Latin in Latin text, and never mix scripts inside one word.

How to write it

How to write the name in each situation it appears in.	
Situation	Write
Full name, English	Aninda Studio
Full name, Bangla	অনিন্দ্য স্টুডিও
Short form, English	Aninda
Short form, Bangla	অনিন্দ্য
Repository name	aninda-studio
The two token packages aninda-studio-tokens	
Domain	anindastudio.com — chosen, and not registered as of 19 August 2026
Sentence case, always. Not ANINDA STUDIO, not Aninda STUDIO. Capitals are harder to read, and in Bangla they do not exist at all, so an all-capitals house style would have nothing to translate into.	

The name is an adjective, not a verb or a noun

This is a convention borrowed from Apple's own trademark guidance, and it is a good one. A name used as a verb or pluralised stops being a name and starts being a generic word, which is how trademarks are lost.

- Correct: *the Aninda Studio design system, an Aninda Studio component*
- Wrong: *Anindas, Aninda'd, we Anindaed the page*

What the name means

Anindya means faultless or beyond reproach. That is a high thing to be called and I do not claim it as a description of the work. It is a family name, given before any of this existed. Chapter 14 is the part of this book that keeps the name honest.

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The verified name strings

Both are proper names. Rule 4 of the Bangla Academy's 2012 spelling rules exempts the names of people, institutions and organisations from the spelling rules outright. Independently of that exemption, both are correctly formed: স্টুডিও takes an initial conjunct under rule 2.9, in the same pattern as সেশন, and a short ই-কার under rule 2.1.

The name in Bangla, long
form and short.

Use	String
-----	--------

Full name	অনিন্দ্য স্টুডিও
-----------	------------------

Short form	অনিন্দ্য
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The Latin form is **aninda**. অনিন্দ্য transliterates as *anindya*; *aninda* is the ordinary everyday Latin spelling of the same name. That is transliteration, not an inconsistency. The English section says the same thing at greater length.

The mark • চিহ্ন

A circle tangent to a stem that overruns downward. One geometry, two weights, and the mistake that produced it.

English

The mark is a circle tangent to a stem. The stem overruns the circle **downward**.



Regular — stroke 9



Heavy — stroke 15

That is the whole idea. A river meeting the sea is a circle of still water with a channel running out of it, and Barishal sits in a delta. The mark is not a letter and is not asked to be one.

The mistake that shaped it, recorded so it is not repeated

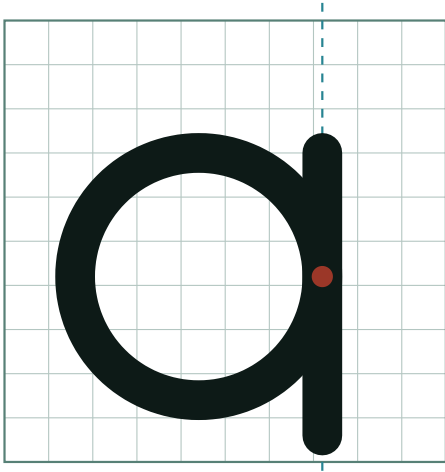
An earlier draft ran the stem **upward** past the top of the circle. Drawn that way, a circle with a vertical stroke rising on its right side is a lowercase **d**. Not something like a *d* — it is a *d*. The studio's Latin name begins with an *a*, so the mark was quietly spelling the wrong letter.

Nothing in the specification caught this. The numbers were fine, the tangency was exact, the proportions were the ones I had chosen. It was found by rendering the thing and looking at it. That is the argument for building the artefact early rather than describing it well, and it is why every stage of this project ends in a rendered file rather than a document.

Turning the overrun downward fixed it. The circle and the stem are otherwise unchanged.

Construction

The mark is drawn on a 100-unit grid. These figures are read from `04_mark/manifest.json` at build time.



The circle, the tangent stem, and the downward overrun. Read from 04_mark/manifest.json.

Read from 04_mark/manifest.json at build time.

Measure	Value
Grid	100 × 100 units
Circle centre	x 44, y 58
Circle radius	28 units
Stem x	72 — the circle's right edge, so the two are tangent
Stem top	y 30 — level with the top of the circle
Stem bottom	y 94 — 8 units below the circle
Clear space	half the mark's own height on all four sides
Safe field inside an icon	90 of 100 units

The stem sits at the exact x of the circle's right edge, so the two are tangent rather than overlapping. The stem starts level with the top of the circle and runs past the bottom of it.

Two weights, one geometry

The rule, in the manifest's own words: stroke 9 at 24 px and above; stroke 15 below.

Weight	Stroke	Use at	Drawn extent	Clear space
Regular	9	24 px and above	65 × 73 units	36.5 units
Heavy	15	below 24 px	71 × 79 units	39.5 units

There is one geometry and two stroke widths. Nothing else changes: the circle does not move, the stem does not move, and the two weights are the same drawing with a heavier pen. A thin stroke disappears at small sizes, so below 24 px the heavy weight takes over. Choose by the size the mark will be seen at, not by taste.

Both files are drawn in `currentColor`, so the mark takes the colour of whatever text it sits in, in any of the four themes and in forced colours.

Clear space

Clear space is half the mark's own height, on all four sides.

It is written as a proportion rather than a fixed measure so that it holds at every size without a table. At the regular weight the drawn mark is 73 units tall on the 100-unit grid, so the clear space is 36.5 units. At the heavy weight the mark is 79 units tall and the clear space is 39.5.

Nothing enters that space. Not a caption, not a rule, not the edge of a photograph, not another logo.

Minimum size

The stroke rule is the minimum-size rule. Below 24 px use the heavy weight. Below 16 px, use the icon rather than the bare mark: the icon has a solid ground behind it and survives where a hairline on a busy background does not.

What not to do

- Do not run the stem upward. That is the *d*, and it is the one mistake this mark has already made.
- Do not add a second colour. The mark is one colour and takes it from its context.
- Do not outline it, add a shadow, or set it on a gradient.
- Do not stretch it. Scale both axes together.
- Do not rotate it.
- Do not redraw it at a stroke width between 9 and 15. There are two weights, and a third one drawn by hand will not match either.
- Do not put the mark and the wordmark at arbitrary distances from each other. Use the lockups, or leave clear space and set them on a shared baseline.

The files

Every mark file this build wrote, with its size read off the file.

File	Size
mark-regular.svg	457 bytes
mark-heavy.svg	457 bytes
wordmark-latin.svg	25 kB
wordmark-bangla.svg	28 kB
tile-web.svg	495 bytes
icon-1024.svg	527 bytes
icon-512.svg	519 bytes
icon-192.svg	507 bytes
icon-apple-1024.svg	545 bytes
icon-apple-1088-watch.svg	512 bytes
icon-apple-1024-dark.svg	509 bytes
icon-apple-1024-mono.svg	478 bytes
icon-android-background-108.svg	258 bytes
icon-android-foreground-108.svg	461 bytes
icon-android-monochrome-108.svg	487 bytes

Every check the mark build ran, and passed

- uharfbuzz present, HarfBuzz 14.3.0
- Bangla wordmark: 16 code points → 11 glyphs, advance 520.1
- Latin wordmark: 13 code points → 13 glyphs, advance 653.2
- conjuncts formed: 15 code points → 11 glyphs
- negative control passed: naive 16 glyphs ≠ shaped 11 glyphs
- icon at stroke 9: mark 65.0×73.0 scaled ×0.9208, worst corner exactly on the 45-unit inscribed circle — the scale is derived from the mark's own diagonal, so this is a fit by construction; what is tested is that all four corners also land inside the 90-unit field
- icon at stroke 15: mark 71.0×79.0 scaled ×0.8473, worst corner exactly on the 45-unit inscribed circle — the scale is derived from the mark's own diagonal, so this is a fit by construction; what is tested is that all four corners also land inside the 90-unit field
- icon at stroke 9 on the 108-unit grid: mark 65.0×73.0 scaled ×0.6752, worst corner exactly on the 33-unit inscribed circle — the scale is derived from the mark's own diagonal, so this is a fit by construction; what is tested is that all four corners also land inside the 66-unit field
- 4 recolourable files carry no root colour and draw in `currentColor`
- tile-web.svg: 4.7% background showing — rounded
- icon-1024.svg: 4.7% background showing — rounded — the web icon
- icon-512.svg: 4.7% background showing — rounded
- icon-192.svg: 4.7% background showing — rounded
- icon-apple-1024.svg: 0.0% background showing — square and fully opaque — Apple, Default

- icon-apple-1088-watch.svg: 0.0% background showing — square and fully opaque — Apple, watchOS
- icon-apple-1024-dark.svg: 0.0% background showing — square and fully opaque — Apple, Dark
- icon-android-background-108.svg: 0.0% background showing — flat and fully opaque — Android background
- icon-apple-1024-mono.svg: 82.6% background showing — no ground; the alpha carries the shape — Apple, Mono
- icon-android-foreground-108.svg: 91.7% background showing — no ground; the alpha carries the shape — Android foreground
- icon-android-monochrome-108.svg: 91.7% background showing — no ground; the alpha carries the shape — Android monochrome
- under a circle inscribed in the frame, icon-1024.svg and icon-apple-1024.svg are the same image in all 65536 pixels — the corner rounding lies entirely outside the circle watchOS and visionOS mask to, and the artwork inside it has not drifted between the two files
- visual parity: the mark fills 59.850x67.216 per cent of the Apple frame and 60.958x68.461 per cent of Android's visible 72 dp viewport — a difference of 1.11 and 1.24 percentage points. The corner shape differs between the platforms by decision; the size does not
- contact sheet: 15 artefacts nested from the same strings written to 04_mark/svg, every caption read out of the artwork

Bangla · বাংলা

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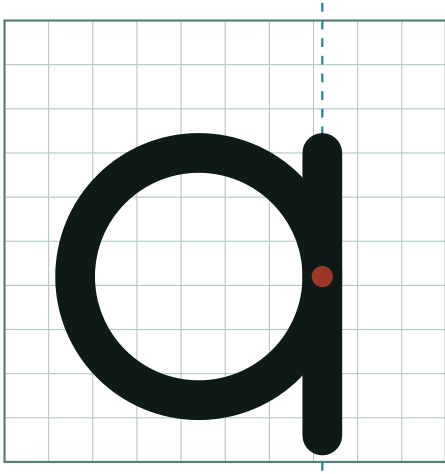
The drawing itself carries no language. The construction figures, the two stroke weights, the clear-space rule and the file list in the English section apply without change, and the numbers in them are read from the same source file for both languages.



Regular — stroke 9



Heavy — stroke 15



The circle, the tangent stem, and the downward overrun. Read from 04_mark/manifest.json.

Read from 04_mark/manifest.json at build time.

Measure	Value
Grid	100 × 100 units
Circle centre	x 44, y 58
Circle radius	28 units
Stem x	72 — the circle's right edge, so the two are tangent
Stem top	y 30 — level with the top of the circle
Stem bottom	y 94 — 8 units below the circle
Clear space	half the mark's own height on all four sides
Safe field inside an icon	90 of 100 units

The rule, in the manifest's own words: stroke 9 at 24 px and above; stroke 15 below.

Weight	Stroke	Use at	Drawn extent	Clear space
Regular	9	24 px and above	65 × 73 units	36.5 units
Heavy	15	below 24 px	71 × 79 units	39.5 units

Icons • আইকন

One rounded icon on every surface, Apple included — and exactly what that trades away.

English



icon-1024.svg

The web icon, rounded



icon-512.svg

Avatars and progressive web apps



icon-192.svg

Progressive web apps



tile-web.svg

Web tile, heavy weight



icon-apple-1024.svg

Apple, Default — square, unmasked



icon-apple-1024-dark.svg

Apple, Dark



icon-apple-1024-mono.svg

Apple, Mono — no ground



icon-apple-1088-watch.svg

Apple, watchOS at 1088



icon-android-background-108.svg

Android, background layer



icon-android-foreground-108.svg

Android, foreground layer



icon-android-monochrome-108.svg

Android, monochrome — the system tints this

The decision

Each platform's own icon geometry is followed. Apple and Android are given square, unmasked artwork and apply their own masks. The web keeps the rounded tile, because a browser will not round a favicon for you. That is the owner's decision, taken on 26 August 2026.

It reverses the opposite decision, taken on 14 August 2026, which was to use one rounded icon on every surface, Apple included. Both are recorded, because a reversed decision that vanishes teaches nobody anything.

Every icon artefact, and the platform each one is for.

File	Which platform it is for
tile-web.svg	The web. Rounded, because a browser will not round a favicon for you
icon-1024.svg	The web. Rounded, because a browser will not round a favicon for you
icon-512.svg	The web. Rounded, because a browser will not round a favicon for you
icon-192.svg	The web. Rounded, because a browser will not round a favicon for you
icon-apple-1024.svg	Apple. Square and unmasked — the system applies its own mask
icon-apple-1088-watch.svg	Apple. Square and unmasked — the system applies its own mask
icon-apple-1024-dark.svg	Apple. Square and unmasked — the system applies its own mask
icon-apple-1024-mono.svg	Apple. Square and unmasked — the system applies its own mask
icon-android-background-108.svg	Android. A layer the launcher composites
icon-android-foreground-108.svg	Android. A layer the launcher composites
icon-android-monochrome-108.svg	Android. A layer the launcher composites

The decision. Each platform's own icon geometry is followed. Apple and Google receive square, unmasked artwork and apply their own masks. The web keeps the rounded tile, because a browser will not round a favicon for you. Owner's decision, 26 August 2026. This kit now ships to two developer accounts. Both platforms ask for unmasked artwork and both derive something from the edges of what they are given: Apple its Liquid Glass specular highlights, Google its own corner mask and drop shadow. Google publishes a figure where Apple does not — a radius of 30 per cent of the icon size, applied by Play. Supplying pre-rounded artwork means both of those follow the wrong geometry, and the cost of that could not be measured outside their renderers. Supplying what each asks for removes the unknown instead of accepting it. Verified against the Apple Human Interface Guidelines and the Google Play icon design specifications, checked 26 August 2026.

What this reversed. On 14 August 2026 the decision here was the opposite: One rounded icon is used on every surface, Apple included. It was reversed on 26 August 2026. This kit now ships to both stores, and both ask for unmasked artwork. The measured half still holds: under a circular mask the rounded icon and the square master were the same image in every pixel. What changed is the judged half and the unknown half — and the unknown one is now removed rather than carried, because each platform is given the geometry it asks for.

Why it was reversed

This kit now ships to two developer accounts, and both platforms ask for the same thing in different words.

Apple asks for square, unmasked artwork. The system applies the mask itself, and it derives the Liquid Glass specular highlights — the moving glints along the edge of an icon — from the edges of the layers you supply. A pre-rounded edge sits inside the mask the system draws, so the highlight follows the wrong geometry. Apple's own wording is that pre-masked artwork "negatively impacts specular highlight effects" and makes edges "look jagged".

Google asks for the same, and is more specific about it. The Play Store icon is a full square. Play applies a corner radius **equivalent to 30% of the icon size** and adds the drop shadow itself, and the specification says not to bake either one in.

That contrast is worth noticing. **Google publishes a number here and Apple does not.** It is the same rule on both platforms, and only one of them can be quoted.

What the old decision accepted, and what has changed about it

The 14 August decision was recorded with three claims, each labelled as a measurement, a judgement, or neither. That structure still holds, and two of the three have moved.

Measured, and still true: on watchOS and visionOS the rounding cost nothing. Both mask to a **circle**, and the corner material a radius removes lies outside a circle inscribed in the same square. `04_mark/build.py` renders the rounded web icon and the square Apple master under that circle and differences the two images pixel by pixel.

Measured. Under a circle inscribed in the frame, `icon-1024.svg` and `icon-apple-1024.svg` are the same image in all 65536 pixels — the corner rounding lies entirely outside the circle watchOS and visionOS mask to, and the artwork inside it has not drifted between the two files.

Judged, not measured: the static difference under the rounded-rectangle mask looked slight. That was a reading of a static render. It could not be a measurement, because Apple publishes no corner radius, so there is no mask to composite against without substituting this kit's own radius for Apple's and then measuring the substitution.

Neither, and this is the one that decided it: the dynamic cost. Liquid Glass highlights are generated at run time by Apple's renderer, from the layer edges, as the device moves. There is no way to reproduce that outside that renderer, so the difference in the moving highlight was never measured and was never known.

The old decision accepted that unknown in exchange for one icon instead of several. The new one removes it instead, by giving each platform the geometry it asks for. Nothing was measured that had not been measured before. What changed is that an unknown no longer has to be carried.

What differs between the platforms, and what does not

The corner shape now differs. The size of the mark does not, and that is measured rather than hoped for.

The two canvases invite a mistake here. Apple's mark sits inside 100 units, all of which are shown. Android's sits inside 108 units, of which only the middle 72 are ever displayed. Comparing the two scales directly would say they are very different and would mean nothing. What matters is the fraction of the **visible** area the mark fills.

Measured. Visual parity: the mark fills 59.850x67.216 per cent of the Apple frame and 60.958x68.461 per cent of Android's visible 72 dp viewport — a difference of 1.11 and 1.24 percentage points. The corner shape differs between the platforms by decision; the size does not.

The build fails if that difference ever exceeds two percentage points.

The Android layers

Android's adaptive icon is three layers on a 108 dp canvas: a background, a foreground, and a monochrome silhouette the system tints to match the wallpaper. The outer 18 dp on each side is reserved for masking and for motion effects, and the middle **66 dp** is the zone no launcher mask may clip. Both figures are Google's.

The monochrome layer matters more than its size suggests. From Android 13 a person can ask for themed icons, and from Android 16 QPR 2 the system generates a monochrome layer for any app that does not supply one. Supplying the shape by hand is better than having it inferred.

One shortfall is recorded rather than designed away. Google asks for a logo of at least 48 dp. At the scale that keeps every inked pixel inside the safe circle, this mark measures **43.89 × 49.29 dp** — so it meets 48 dp on its long axis and not on its short one. Fitting the 66 dp square instead would meet both, and would put ink outside the circle a round launcher mask leaves. The circle wins, because a clipped mark is a worse fault than a narrow one.

The corner radius is this kit's own number

The web tile rounds by 24% of its width. The build reads that numeral from this system's own radius-hero token, so it is not typed anywhere.

Be precise about what that means. radius-hero is **24 px**; the tile rounds by 24 units per 100 on its grid, which is 245.8 px at 1024 px. The number is reused on purpose, so the tile's corner belongs to the same family as every other rounded corner in the kit — an echo, not a unit conversion.

Apple publishes no corner radius, no percentage, and does not use the word "squircle" anywhere in current guidance. The widely circulated community figure of 22.37% with roughly 60% corner smoothing describes a superellipse — a rounded shape whose curvature varies continuously —

which a single radius value cannot describe at all. Any radius quoted in this kit is this kit's, and is not claimed to be Apple's.

The safe field

On the 100-unit grid the mark sits inside a 90-unit field, centred. That is this kit's own rule. **Apple publishes no numeric safe zone** for iOS, iPadOS or macOS; the guidance is qualitative. Stating a percentage and attributing it to Apple would be a fabrication.

On the 108-unit Android grid the field is Google's 66 dp, not this kit's 90. The same placement rule serves both: fit the mark's diagonal inside the inscribed circle, because a circle is the tightest mask any platform applies.

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icon-1024.svg

The web icon, rounded



icon-512.svg

Avatars and progressive web apps



icon-192.svg

Progressive web apps



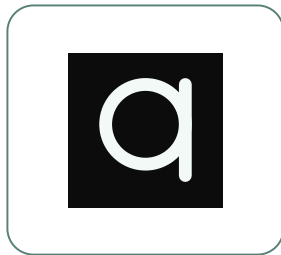
tile-web.svg

Web tile, heavy weight



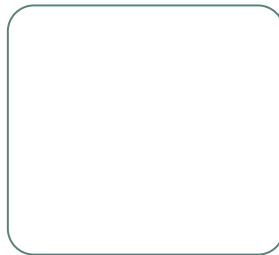
icon-apple-1024.svg

Apple, Default — square, unmasked



icon-apple-1024-dark.svg

Apple, Dark



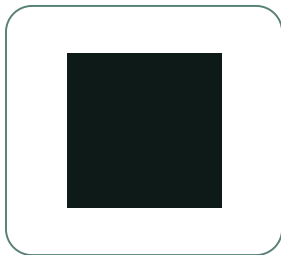
icon-apple-1024-mono.svg

Apple, Mono — no ground



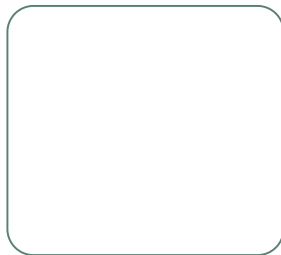
icon-apple-1088-watch.svg

Apple, watchOS at 1088



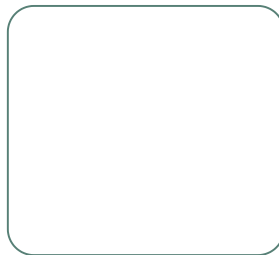
icon-android-background-108.svg

Android, background layer



icon-android-foreground-108.svg

Android, foreground layer



icon-android-monochrome-108.svg

Android, monochrome — the system tints this

Every icon artefact, and the platform each one is for.

File	Which platform it is for
tile-web.svg	The web. Rounded, because a browser will not round a favicon for you
icon-1024.svg	The web. Rounded, because a browser will not round a favicon for you
icon-512.svg	The web. Rounded, because a browser will not round a favicon for you
icon-192.svg	The web. Rounded, because a browser will not round a favicon for you
icon-apple-1024.svg	Apple. Square and unmasked — the system applies its own mask
icon-apple-1088-watch.svg	Apple. Square and unmasked — the system applies its own mask
icon-apple-1024-dark.svg	Apple. Square and unmasked — the system applies its own mask
icon-apple-1024-mono.svg	Apple. Square and unmasked — the system applies its own mask
icon-android-background-108.svg	Android. A layer the launcher composites
icon-android-foreground-108.svg	Android. A layer the launcher composites
icon-android-monochrome-108.svg	Android. A layer the launcher composites

The decision. Each platform's own icon geometry is followed. Apple and Google receive square, unmasked artwork and apply their own masks. The web keeps the rounded tile, because a browser will not round a favicon for you. Owner's decision, 26 August 2026. This kit now ships to two developer accounts. Both platforms ask for unmasked artwork and both derive something from the edges of what they are given: Apple its Liquid Glass specular highlights, Google its own corner mask and drop shadow. Google publishes a figure where Apple does not — a radius of 30 per cent of the icon size, applied by Play. Supplying pre-rounded artwork means both of those follow the wrong geometry, and the cost of that could not be measured outside their renderers. Supplying what each asks for removes the unknown instead of accepting it. Verified against the Apple Human Interface Guidelines and the Google Play icon design specifications, checked 26 August 2026.

What this reversed. On 14 August 2026 the decision here was the opposite: One rounded icon is used on every surface, Apple included. It was reversed on 26 August 2026. This kit now ships to both stores, and both ask for unmasked artwork. The measured half still holds: under a circular mask the rounded icon and the square master were the same image in every pixel. What changed is the judged half and the unknown half — and the unknown one is now removed rather than carried, because each platform is given the geometry it asks for.

Colour • রং

Every ramp, every role and every measured ratio in four themes, read out of the token files.

English

Three terms, first

Contrast ratio

How far apart two colours are in brightness, written as a ratio. Black on white is 21:1; two colours you cannot tell apart are 1:1.

WCAG AA

The middle of the three conformance levels in the Web Content Accessibility Guidelines: 4.5:1 for normal text and 3:1 for borders, icons and controls. This system is held to it.

WCAG AAA

The strictest level: 7:1 for normal text. This system reaches it in places, and the high contrast themes are generated against it rather than patched towards it.

WCAG defines no AAA level for non-text contrast. Criterion 1.4.11 has a single requirement of 3:1 and no enhanced tier above it. So a border measured at 3.9:1 has *fully met* 1.4.11. It is not an AA-only compromise waiting to be improved, and nothing in this book will describe it as one.

The direction

Estuary — মোহনা

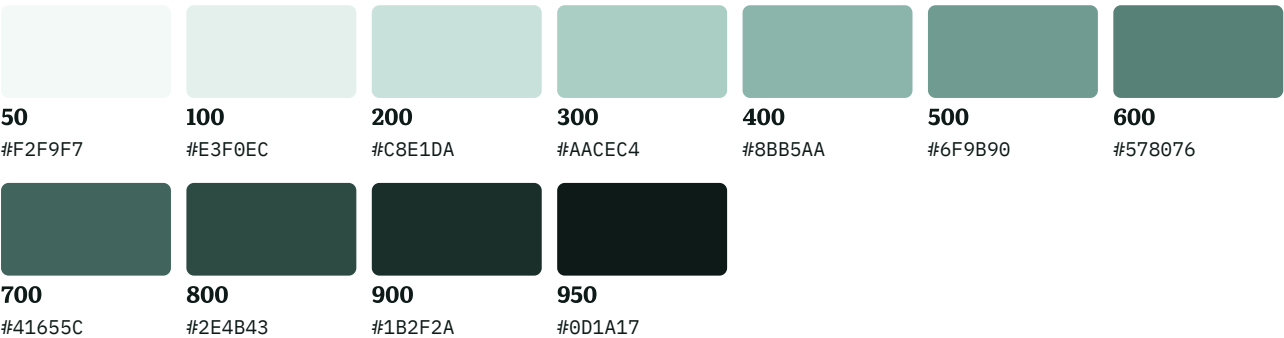
Quiet editorial competence. A deep green-teal ground against a bone paper, with the warmth coming from the paper rather than from a second expressive colour. Barishal sits in a river delta; an estuary is where fresh water meets salt, which is also what a studio serving two audiences in two scripts is doing. This is Direction A: the existing draft, corrected. One correction is substantive — the draft declares no accent at all, and a system whose accent equals its text colour cannot express a link, a selected row, or a primary action. A cyan-teal accent is therefore introduced: a near neighbour of the ground, discriminable from it, and still honouring 'no second expressive colour'.

The six ramps

Each ramp is computed in OKLCH — a colour space where a fixed change in lightness looks like the same change to the eye at every hue — then mapped into sRGB. Each rung is at least 0.9 ΔE2000 from the one before it, which is the smallest difference a person reliably sees. The 11 steps run 50 to 950.

Estuary — মোহনা / ground

The ground, and the source of every surface tint and every text role.

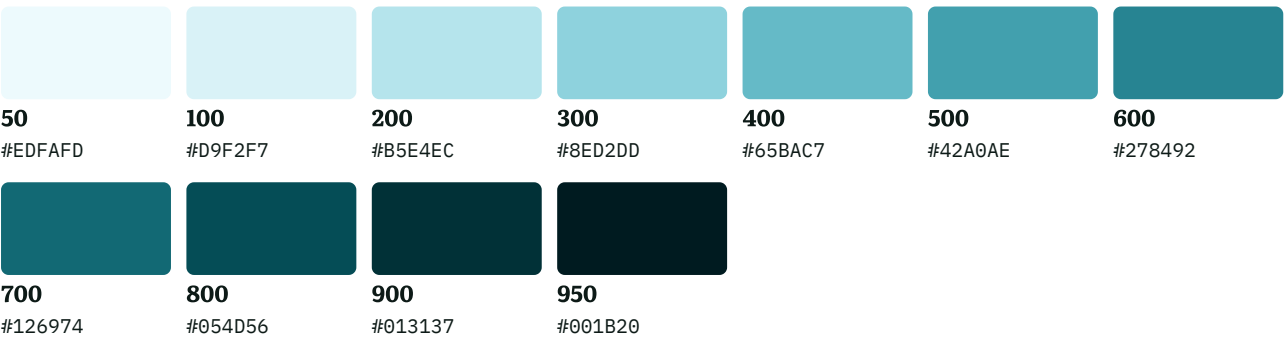


How the Estuary ramp is generated.

Hue (OKLCH)	Chroma ceiling	Anchor	Anchor step	Kind
176.896°	0.05127	#0C3A31	900	brand

Tidewater — জোয়ার / accent

A cyan-teal one hue-step off the ground. Carries links, focus and the primary action.

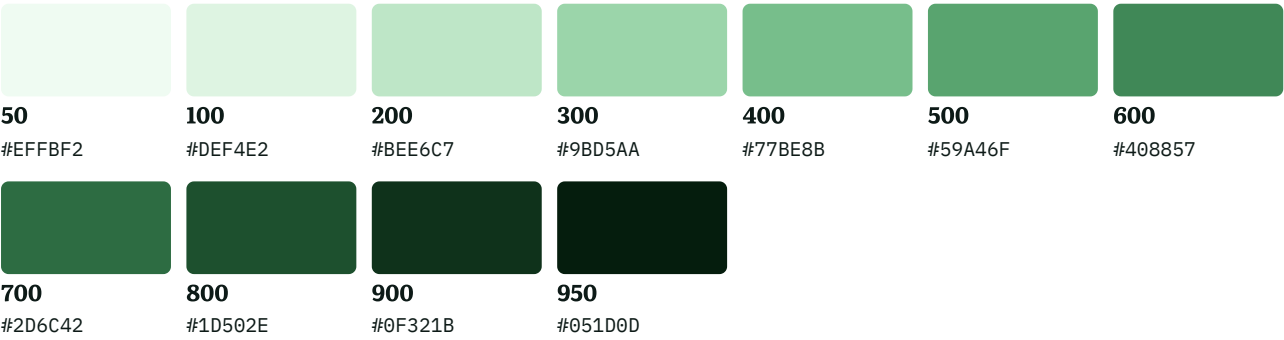


How the Tidewater ramp is generated.

Hue (OKLCH)	Chroma ceiling	Anchor	Anchor step	Kind
208.996°	0.0899	#0E7C8A	600	brand

Silt — পলি / success

Silt (পলি) — semantic

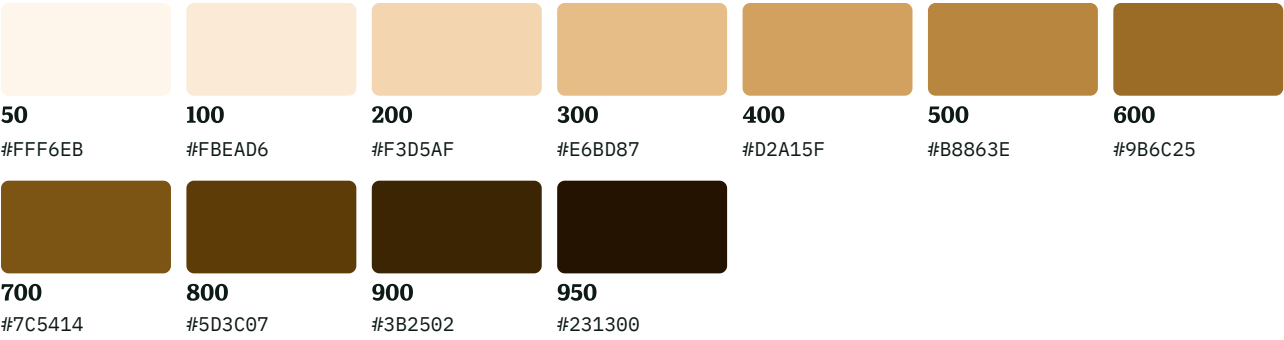


How the Silt ramp is generated.

Hue (OKLCH)	Chroma ceiling	Anchor	Anchor step	Kind
152.008°	0.10913	#1B6B3A	700	semantic

Kans — কাশ / warning

Kans (কাশ) — semantic

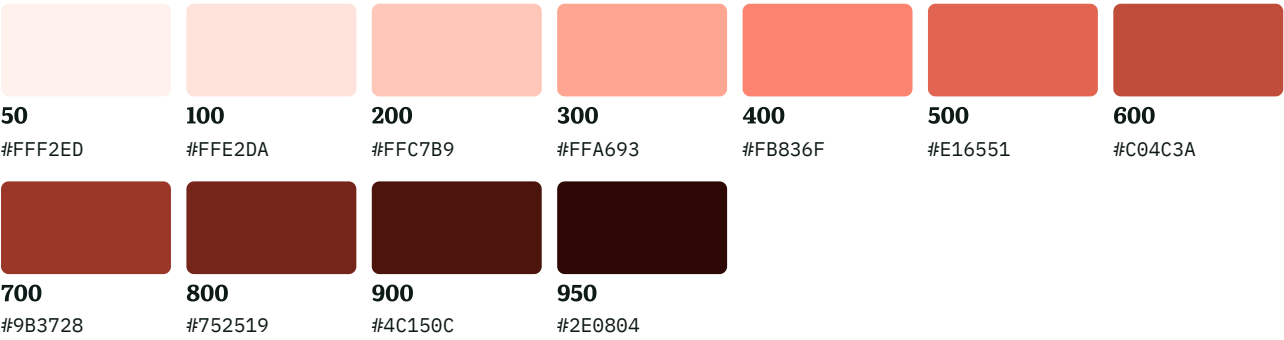


How the Kans ramp is generated.

Hue (OKLCH)	Chroma ceiling	Anchor	Anchor step	Kind
73.298°	0.108	#8A5A00	700	semantic

Laterite — লাল মাটি / danger

Laterite (লাল মাটি) — semantic

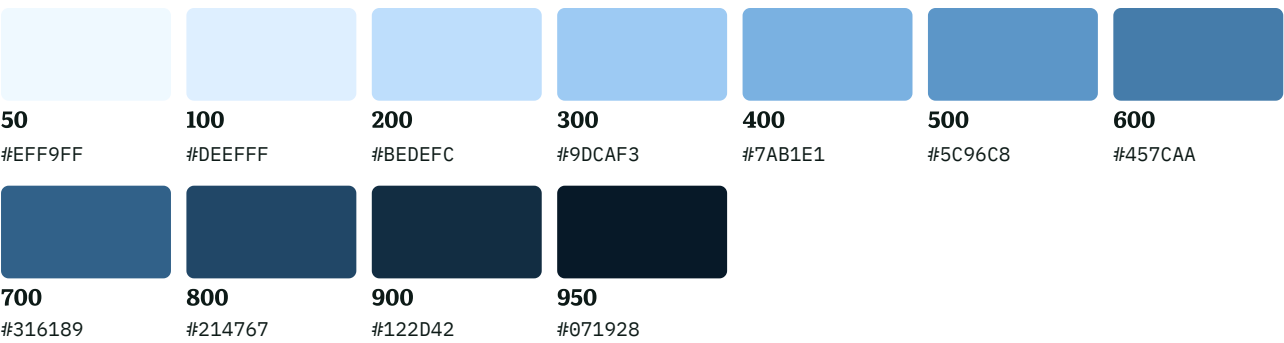


How the Laterite ramp is generated.

Hue (OKLCH)	Chroma ceiling	Anchor	Anchor step	Kind
31.189°	0.15914	#A8301F	700	semantic

Monsoon — বর্ষা / info

Monsoon (বর্ষা) — semantic



How the Monsoon ramp is generated.

Hue (OKLCH)	Chroma ceiling	Anchor	Anchor step	Kind
245.674°	0.0962	#235E8C	700	semantic

The four themes

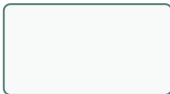
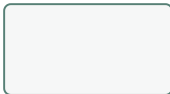
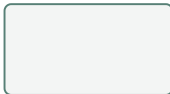
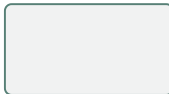
Not four skins over one palette. Each theme was generated against its own contrast target, which is why the high contrast pair is a different set of colours rather than the same colours pushed further apart.

Light — আলো

What the Light theme is measured against.

Setting	Value
Polarity	light
High contrast	no
Text target	4.50:1
Non-text target	3.00:1

Surfaces

						
lowest #FDFFFE	low #FBFCFC	base #F8FAF9	high #F6F7F7	highest #F3F5F4	dim #F1F2F2	bright #FFFFFF
luminance 0.9956	luminance 0.9716	luminance 0.9517	luminance 0.9283	luminance 0.9089	luminance 0.8862	luminance 1.0000

Roles

Every role in the Light theme, with the contrast ratio it was measured at and the criterion it was measured against.

Role	Value	Ramp step	Kind	Required	Measured	Worst case ±1 bit	Hardest surface	Level	Criterion
● ink	#0D1A17	ground 950	text	4.50:1	15.90:1	15.61:1	dim	AAA	WCAG 2.2 1.4.3
● ink-muted	#41655C	ground 700	text	4.50:1	5.78:1	5.64:1	dim	AA	WCAG 2.2 1.4.3
● line	#578076	ground 600	nontext	3.00:1	3.94:1	3.85:1	dim	meets 1.4.11	WCAG 2.2 1.4.11
● accent	#126974	accent 700	text	4.50:1	5.68:1	5.55:1	dim	AA	WCAG 2.2 1.4.3
● accent-edge	#278492	accent 600	nontext	3.00:1	3.90:1	3.81:1	dim	meets 1.4.11	WCAG 2.2 1.4.11
● accent-hover	#054D56	accent 800	fill	4.50:1	9.50:1	9.28:1	label (surface.lowest)	AAA	WCAG 2.2 1.4.3
○ accent-on	#FDFFFE	surface lowest	on-fill	4.50:1	6.34:1	6.20:1	accent	AA	WCAG 2.2 1.4.3
● focus-ring	#278492	accent 600	nontext	3.00:1	3.90:1	3.81:1	dim	meets 1.4.11	WCAG 2.2 1.4.11
● status-success	#2D6C42	success 700	text	4.50:1	5.61:1	5.48:1	dim	AA	WCAG 2.2 1.4.3
● status-warning	#7C5414	warning 700	text	4.50:1	5.97:1	5.83:1	dim	AA	WCAG 2.2 1.4.3
● status-danger	#9B3728	danger 700	text	4.50:1	6.30:1	6.16:1	dim	AA	WCAG 2.2 1.4.3
● status-info	#316189	info 700	text	4.50:1	5.85:1	5.71:1	dim	AA	WCAG 2.2 1.4.3

Every ink and line role against every one of the seven surfaces

Every role that sits ON a surface, against every surface in the Light theme. Each cell is a measured contrast ratio. accent-hover and accent-on are absent because they are grounds that carry text, so they are measured against the labels on them — see the Hardest surface column above.

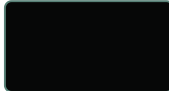
Role	lowest	low	base	high	highest	dim	bright
ink	17.76:1	17.36:1	17.02:1	16.62:1	16.29:1	15.90:1	17.84:1
ink-muted	6.45:1	6.30:1	6.18:1	6.04:1	5.92:1	5.78:1	6.48:1
line	4.40:1	4.30:1	4.21:1	4.12:1	4.03:1	3.94:1	4.42:1
accent	6.34:1	6.19:1	6.07:1	5.93:1	5.81:1	5.68:1	6.37:1
accent-edge	4.36:1	4.26:1	4.17:1	4.07:1	3.99:1	3.90:1	4.37:1
focus-ring	4.36:1	4.26:1	4.17:1	4.07:1	3.99:1	3.90:1	4.37:1
status-success	6.27:1	6.13:1	6.01:1	5.87:1	5.75:1	5.61:1	6.30:1
status-warning	6.67:1	6.52:1	6.39:1	6.24:1	6.12:1	5.97:1	6.70:1
status-danger	7.04:1	6.88:1	6.74:1	6.59:1	6.46:1	6.30:1	7.07:1
status-info	6.53:1	6.38:1	6.26:1	6.11:1	5.99:1	5.85:1	6.56:1

Dark — অন্ধকার

What the Dark theme is measured against.

Setting	Value
Polarity	dark
High contrast	no
Text target	4.50:1
Non-text target	3.00:1

Surfaces

						
lowest #0B0C0B	low #0E100F	base #111212	high #111312	highest #121313	dim #060707	bright #121413
luminance 0.0036	luminance 0.0050	luminance 0.0060	luminance 0.0063	luminance 0.0064	luminance 0.0021	luminance 0.0068

Roles

Every role in the Dark theme, with the contrast ratio it was measured at and the criterion it was measured against.

Role	Value	Ramp step	Kind	Required	Measured	Worst case ±1 bit	Hardest surface	Level	Criterion
 ink	#F2F9F7	ground 50	text	4.50:1	17.33:1	17.03:1	bright	AAA	WCAG 2.2 1.4.3
 ink-muted	#6F9B90	ground 500	text	4.50:1	5.96:1	5.84:1	bright	AA	WCAG 2.2 1.4.3
 line	#578076	ground 600	nontext	3.00:1	4.19:1	4.09:1	bright	meets 1.4.11	WCAG 2.2 1.4.11
 accent	#42A0AE	accent 500	text	4.50:1	6.05:1	5.93:1	bright	AA	WCAG 2.2 1.4.3
 accent-edge	#278492	accent 600	nontext	3.00:1	4.23:1	4.14:1	bright	meets 1.4.11	WCAG 2.2 1.4.11
 accent-hover	#65BAC7	accent 400	fill	4.50:1	8.77:1	8.62:1	label (surface.lowest)	AAA	WCAG 2.2 1.4.3
 accent-on	#0B0C0B	surface lowest	on-fill	4.50:1	5.77:1	5.66:1	danger	AA	WCAG 2.2 1.4.3
 focus-ring	#278492	accent 600	nontext	3.00:1	4.23:1	4.14:1	bright	meets 1.4.11	WCAG 2.2 1.4.11
 status-success	#59A46F	success 500	text	4.50:1	6.13:1	6.01:1	bright	AA	WCAG 2.2 1.4.3
 status-warning	#B8863E	warning 500	text	4.50:1	5.74:1	5.62:1	bright	AA	WCAG 2.2 1.4.3
 status-danger	#E16551	danger 500	text	4.50:1	5.45:1	5.33:1	bright	AA	WCAG 2.2 1.4.3
 status-info	#5C96C8	info 500	text	4.50:1	5.86:1	5.74:1	bright	AA	WCAG 2.2 1.4.3

Every ink and line role against every one of the seven surfaces

Every role that sits ON a surface, against every surface in the Dark theme. Each cell is a measured contrast ratio. `accent-hover` and `accent-on` are absent because they are grounds that carry text, so they are measured against the labels on them — see the `Hardest surface` column above.

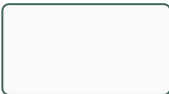
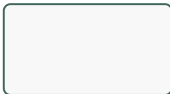
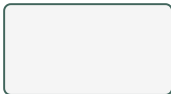
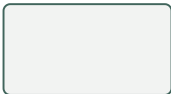
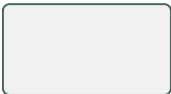
Role	lowest	low	base	high	highest	dim	bright
ink	18.35:1	17.89:1	17.58:1	17.47:1	17.43:1	18.89:1	17.33:1
ink-muted	6.31:1	6.15:1	6.05:1	6.01:1	6.00:1	6.50:1	5.96:1
line	4.44:1	4.32:1	4.25:1	4.22:1	4.21:1	4.57:1	4.19:1
accent	6.41:1	6.25:1	6.14:1	6.10:1	6.09:1	6.60:1	6.05:1
accent-edge	4.48:1	4.37:1	4.29:1	4.27:1	4.26:1	4.61:1	4.23:1
focus-ring	4.48:1	4.37:1	4.29:1	4.27:1	4.26:1	4.61:1	4.23:1
status-success	6.50:1	6.33:1	6.22:1	6.19:1	6.17:1	6.69:1	6.13:1
status-warning	6.08:1	5.93:1	5.82:1	5.79:1	5.78:1	6.26:1	5.74:1
status-danger	5.77:1	5.62:1	5.52:1	5.49:1	5.48:1	5.94:1	5.45:1
status-info	6.21:1	6.05:1	5.94:1	5.91:1	5.90:1	6.39:1	5.86:1

High contrast, light — বেশি কনট্রাস্ট, আলো

What the High contrast, light theme is measured against.

Setting	Value
Polarity	light
High contrast	yes
Text target	7.00:1
Non-text target	4.50:1

Surfaces

						
lowest	low	base	high	highest	dim	bright
#FCFDFC	#FAFAFA	#F7F8F7	#F5F5F5	#F2F3F2	#F1F1F1	#FFFFFF
luminance	luminance	luminance	luminance	luminance	luminance	luminance
0.9797	0.9560	0.9362	0.9131	0.8939	0.8796	1.0000

Roles

Every role in the High contrast, light theme, with the contrast ratio it was measured at and the criterion it was measured against.

Role	Value	Ramp step	Kind	Required	Measured	Worst case ±1 bit	Hardest surface	Level	Criterion
● ink	#0D1A17	ground 950	text	7.00:1	15.79:1	15.50:1	dim	AAA	WCAG 2.2 1.4.6
● ink-muted	#2E4B43	ground 800	text	7.00:1	8.44:1	8.23:1	dim	AAA	WCAG 2.2 1.4.6
● line	#41655C	ground 700	nontext	4.50:1	5.74:1	5.60:1	dim	meets policy	policy, above WCAG 2.2 1.4.11 — WCAG defines no AAA level for non-text contrast
● accent	#054D56	accent 800	text	7.00:1	8.44:1	8.24:1	dim	AAA	WCAG 2.2 1.4.6
● accent-edge	#126974	accent 700	nontext	4.50:1	5.64:1	5.51:1	dim	meets policy	policy, above WCAG 2.2 1.4.11 — WCAG defines no AAA level for non-text contrast
● accent-hover	#013137	accent 900	fill	7.00:1	13.77:1	13.47:1	label (surface.lowest)	AAA	WCAG 2.2 1.4.6
○ accent-on	#FCFDFC	surface lowest	on-fill	7.00:1	9.35:1	9.14:1	accent	AAA	WCAG 2.2 1.4.6
● focus-ring	#126974	accent 700	nontext	4.50:1	5.64:1	5.51:1	dim	meets policy	policy, above WCAG 2.2 1.4.11 — WCAG defines no AAA level for non-text contrast
● status-success	#1D502E	success 800	text	7.00:1	8.30:1	8.11:1	dim	AAA	WCAG 2.2 1.4.6
● status-warning	#5D3C07	warning 800	text	7.00:1	8.79:1	8.58:1	dim	AAA	WCAG 2.2 1.4.6
● status-danger	#752519	danger 800	text	7.00:1	9.14:1	8.93:1	dim	AAA	WCAG 2.2 1.4.6
● status-info	#214767	info 800	text	7.00:1	8.60:1	8.40:1	dim	AAA	WCAG 2.2 1.4.6

Every ink and line role against every one of the seven surfaces

Every role that sits ON a surface, against every surface in the High contrast, light theme. Each cell is a measured contrast ratio. accent-hover and accent-on are absent because they are grounds that carry text, so they are measured against the labels on them — see the Hardest surface column above.

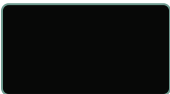
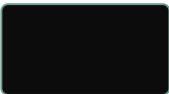
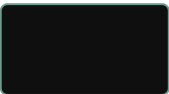
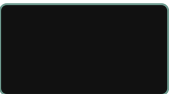
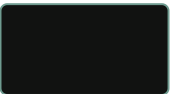
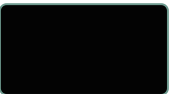
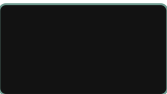
Role	lowest	low	base	high	highest	dim	bright
ink	17.49:1	17.09:1	16.76:1	16.36:1	16.04:1	15.79:1	17.84:1
ink-muted	9.35:1	9.13:1	8.95:1	8.74:1	8.57:1	8.44:1	9.53:1
line	6.36:1	6.21:1	6.09:1	5.94:1	5.83:1	5.74:1	6.48:1
accent	9.35:1	9.14:1	8.96:1	8.75:1	8.57:1	8.44:1	9.54:1
accent-edge	6.24:1	6.10:1	5.98:1	5.84:1	5.72:1	5.64:1	6.37:1
focus-ring	6.24:1	6.10:1	5.98:1	5.84:1	5.72:1	5.64:1	6.37:1
status-success	9.20:1	8.99:1	8.81:1	8.60:1	8.43:1	8.30:1	9.38:1
status-warning	9.74:1	9.51:1	9.33:1	9.11:1	8.93:1	8.79:1	9.93:1
status-danger	10.12:1	9.89:1	9.69:1	9.46:1	9.28:1	9.14:1	10.32:1
status-info	9.53:1	9.31:1	9.12:1	8.91:1	8.73:1	8.60:1	9.71:1

High contrast, dark — বেশি কনট্রাস্ট, অন্ধকার

What the High contrast, dark theme is measured against.

Setting	Value
Polarity	dark
High contrast	yes
Text target	7.00:1
Non-text target	4.50:1

Surfaces

						
lowest	low	base	high	highest	dim	bright
#070807	#0C0C0C	#0E0F0E	#111111	#111211	#030303	#121212
luminance	luminance	luminance	luminance	luminance	luminance	luminance
0.0023	0.0037	0.0047	0.0056	0.0059	0.0009	0.0060

Roles

Every role in the High contrast, dark theme, with the contrast ratio it was measured at and the criterion it was measured against.

Role	Value	Ramp step	Kind	Required	Measured	Worst case ±1 bit	Hardest surface	Level	Criterion
○ ink	#F2F9F7	ground 50	text	7.00:1	17.55:1	17.25:1	bright	AAA	WCAG 2.2 1.4.6
● ink-muted	#8BB5AA	ground 400	text	7.00:1	8.29:1	8.12:1	bright	AAA	WCAG 2.2 1.4.6
● line	#6F9B90	ground 500	nontext	4.50:1	6.04:1	5.91:1	bright	meets policy	policy, above WCAG 2.2 1.4.11 — WCAG defines no AAA level for non-text contrast
● accent	#65BAC7	accent 400	text	7.00:1	8.39:1	8.23:1	bright	AAA	WCAG 2.2 1.4.6
● accent-edge	#42A0AE	accent 500	nontext	4.50:1	6.13:1	6.01:1	bright	meets policy	policy, above WCAG 2.2 1.4.11 — WCAG defines no AAA level for non-text contrast
● accent-hover	#8ED2DD	accent 300	fill	7.00:1	11.86:1	11.67:1	label (surface.lowest)	AAA	WCAG 2.2 1.4.6
● accent-on	#070807	surface lowest	on-fill	7.00:1	8.19:1	8.06:1	danger	AAA	WCAG 2.2 1.4.6
● focus-ring	#42A0AE	accent 500	nontext	4.50:1	6.13:1	6.01:1	bright	meets policy	policy, above WCAG 2.2 1.4.11 — WCAG defines no AAA level for non-text contrast
● status-success	#77BE8B	success 400	text	7.00:1	8.49:1	8.33:1	bright	AAA	WCAG 2.2 1.4.6
● status-warning	#D2A15F	warning 400	text	7.00:1	8.03:1	7.88:1	bright	AAA	WCAG 2.2 1.4.6
● status-danger	#FB836F	danger 400	text	7.00:1	7.65:1	7.51:1	bright	AAA	WCAG 2.2 1.4.6
● status-info	#7AB1E1	info 400	text	7.00:1	8.21:1	8.05:1	bright	AAA	WCAG 2.2 1.4.6

Every ink and line role against every one of the seven surfaces

Every role that sits ON a surface, against every surface in the High contrast, dark theme. Each cell is a measured contrast ratio. accent-hover and accent-on are absent because they are grounds that carry text, so they are measured against the labels on them — see the Hardest surface column above.

Role	lowest	low	base	high	highest	dim	bright
ink	18.79:1	18.32:1	17.99:1	17.69:1	17.59:1	19.32:1	17.55:1
ink-muted	8.87:1	8.65:1	8.49:1	8.35:1	8.30:1	9.12:1	8.29:1
line	6.46:1	6.30:1	6.19:1	6.08:1	6.05:1	6.65:1	6.04:1
accent	8.98:1	8.76:1	8.60:1	8.45:1	8.41:1	9.23:1	8.39:1
accent-edge	6.56:1	6.40:1	6.28:1	6.18:1	6.14:1	6.75:1	6.13:1
focus-ring	6.56:1	6.40:1	6.28:1	6.18:1	6.14:1	6.75:1	6.13:1
status-success	9.10:1	8.87:1	8.71:1	8.56:1	8.51:1	9.35:1	8.49:1
status-warning	8.60:1	8.39:1	8.24:1	8.10:1	8.05:1	8.84:1	8.03:1
status-danger	8.19:1	7.99:1	7.85:1	7.71:1	7.67:1	8.43:1	7.65:1
status-info	8.79:1	8.57:1	8.42:1	8.28:1	8.23:1	9.04:1	8.21:1

Forced colours

Forced colours is the mode where the operating system replaces every colour on the page with its own. Every brand value gives way. A hex that survives this mode has defeated it, which is why the token file maps each role to a system colour keyword instead.

Forced-colors mode: which system keyword each token yields to, from forced-colors.map.json.

Token	System colour it becomes
color.surface.base	Canvas
color.surface.lowest	Canvas
color.surface.low	Canvas
color.surface.high	Canvas
color.surface.highest	Canvas
color.surface.dim	Canvas
color.surface.bright	Canvas
color.ink.default	CanvasText
color.ink.muted	CanvasText
color.line.default	CanvasText
color.accent.default	LinkText
color.accent.edge	CanvasText
color.accent.hover	ButtonFace
color.accent.on	Canvas
color.focus.ring	Highlight
color.status.success	CanvasText
color.status.warning	CanvasText
color.status.danger	CanvasText
color.status.info	CanvasText

- Every brand colour must be overridden. A hex that survives forced-colors mode defeats the whole point of it.
- forced-color-adjust: none is forbidden except where explicitly allow-listed with a stated reason.
- Because status colours all resolve to CanvasText, nothing may rely on colour alone — every state carries a glyph and a word regardless.
- GrayText is reserved for roles that are genuinely disabled, and this map assigns it to none. CSS Color 4 defines it normatively as disabled text, so using it for a live role teaches a high-contrast reader that live content is inactive — and WCAG exempts inactive components from contrast requirements, which would hand away a measured guarantee for nothing. color.ink.muted was mapped to it and paints subtitles, toast bodies and empty-state messages.

Forced-colors mode cannot be expressed in DTCG. Its values are CSS system colour keywords supplied by the operating system — they have no colour space, no components and no hex, and DTCG's thirteen types include nothing that fits. Bending them into a colour token would be a lie about what they are, so this file sits deliberately outside the DTCG tree.

How every figure above was produced

How every colour figure in this chapter was produced.

Step	How
Library	coloraide
Contrast method	wcag21
Perceptual difference	ΔE_{2000}
Ramp space	oklch
Gamut mapping	oklch-chroma

Every ratio is measured on the rounded 8-bit hex, then re-measured with each channel of both colours nudged by ± 1 . The published worst_case_lsb is the lowest of those results.

What the worst-case column means. A colour written as a hex is rounded to 8 bits per channel. Nudging every channel of both colours by one bit, in the direction that hurts, gives the worst ratio the pair can actually produce on a screen. That figure is published beside the measured one. Where the two differ, trust the worst case.

Dynamic colour: this kit holds its own

Dynamic colour derives an app palette from the user's wallpaper and is opt-in per app: a developer must call `applyToActivitiesIfAvailable()`. There are two routes: hold brand colours static, or harmonise them towards the user's palette using `HarmonizedColors`.

This kit takes the first. Brand colours stay static, and `HarmonizedColors` is not used. The reason is the whole point of the colour engine: every pair in this system was measured, and the figure

published is the worst case under a one-bit perturbation. A palette shifted towards a wallpaper at run time has not been measured against anything, so every ratio in this book would become an estimate.

That is a trade, and it is worth naming what it costs: on Android a reader who has set a wallpaper palette will see this system ignore it.

Source: Material 3 specification and Compose Material 3 — `m3.material.io` and `developer.android.com/jetpack/androidx/releases/compose-material3`, checked 14 August 2026. The engine behind the other route is `material-color-utils`, built on HCT — hue, chroma and tone, derived from CAM16 and CIE L* — last pushed 10 August 2026.

Bangla · বাংলা

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The six colour names

All six are brand names. Rule 4 of the Bangla Academy's 2012 rules exempts names from the spelling rules, and each is in any case spelled correctly — পলি takes a short ই-কার under rule 2.1, and বর্ষা keeps its ষ as an unchanged তৎসম under rule 1.1.

The six colour family names in Bangla, each with the ramp it anchors.

Bangla	English	Anchor	Role
মোহনা	Estuary	#0C3A31	ground
জোয়ার	Tidewater	#0E7C8A	accent
পলি	Silt	#1B6B3A	success
কাশ	Kans	#8A5A00	warning
লাল মাটি	Laterite	#A8301F	danger
বর্ষা	Monsoon	#235E8C	info

The four themes

The four theme names in Bangla, and how each string stands.

Bangla	English	Standing
আলো	Light	Verified
অন্ধকার	Dark	Verified
বেশি কনট্রাস্ট, আলো	High contrast, light	Verified th-3 (বেশি কনট্রাস্ট) joined to verified th-1 (আলো) with the same comma the English label uses. Comma is ordinary Western punctuation in Bangla — W3C Bengali orthography notes.
বেশি কনট্রাস্ট, অন্ধকার	High contrast, dark	Verified th-3 joined to verified th-2, same construction as theme.hc-light.

The draft Bangla for high contrast was উচ্চ বসাদৃশ্য, and the review rejected it on two grounds. বসাদৃশ্য means dissimilarity — an abstract comparison between two things — and not the tonal contrast of a display; and উচ্চ is a তৎসম register that clashes with the plain voice used everywhere else. বেশি কনট্রাস্ট is the recommended form, licensed as a loanword by rule 2.6.

The ramps

The numbers do not change between languages. Every ramp, every measured ratio and every proof is in the English section, read from the same token files.

Type • হ্রফ

Two scripts, one scale, a measured multiplier for Bangla and a floor it never goes below.

English

Three families

The three families, read from the font files themselves.

Family	Version	Licence	Variable axes	Reserved Font Name	Subset
Literata	Version 3.103;gftools[0.9.29]	SIL OFL 1.1	opsz 7–72, wght 200–900	no	82 kB
Noto Serif Bengali	Version 3.000	SIL OFL 1.1	wght 100–900, width 62.5–100	no	111 kB
Aninda Mono	Version 2.3	SIL OFL 1.1	none	yes	10 kB

The mono face is IBM Plex Mono, subset and **renamed**. IBM Plex carries a Reserved Font Name, and subsetting counts as modification under clause 3 of the Open Font Licence, so the subset cannot keep the name. Chapter 13 sets out why the other two keep theirs.

The scale

One ratio of 1.333 — a perfect fourth — from caption to display. Sizes are in rem, so they follow the reader's own text-size setting. The pixel column assumes the browser default of 16 px.

The type scale in rem and at a 16 px root, with the measured Bangla multiplier for each step.

Token	Size	At 16 px root	Bangla multiplier	Bangla size
--as-text-caption	0.7502 rem	12.0 px	×0.815	12.0 px (clamped up from 9.8)
--as-text-body	1 rem	16.0 px	×0.816	13.1 px
--as-text-lead	1.333 rem	21.3 px	×0.816	17.4 px
--as-text-h3	1.7769 rem	28.4 px	×0.817	23.2 px
--as-text-h2	2.3686 rem	37.9 px	×0.817	31.0 px
--as-text-h1	3.1573 rem	50.5 px	×0.822	41.5 px
--as-text-display	4.2087 rem	67.3 px	×0.825	55.6 px

Why Bangla needs a multiplier at all

Bangla and Latin do not look the same size at the same size. Bangla's reading height — the distance from the baseline to the মাত্রা, the headline stroke running along the top of the letters — sits at about

0.62 em. Latin's reading height, its x-height, sits at about 0.51 em. Setting both at 16 px makes the Bangla look a fifth larger, so the Bangla is multiplied down until the two measured heights match.

Where the Bangla multiplier comes from: two measured heights at each nominal size.

Nominal size	Literata x-height	Noto Serif Bengali baseline to মাত্রা	Bangla appears larger by	Multiplier
11 px	5.603 px (0.5094 em)	6.842 px (0.6220 em)	×1.2210	×0.819
12 px	6.084 px (0.5070 em)	7.464 px (0.6220 em)	×1.2268	×0.815
16 px	8.117 px (0.5073 em)	9.952 px (0.6220 em)	×1.2260	×0.816
28 px	14.234 px (0.5083 em)	17.416 px (0.6220 em)	×1.2236	×0.817
56 px	28.629 px (0.5112 em)	34.832 px (0.6220 em)	×1.2167	×0.822
100 px	51.300 px (0.5130 em)	62.200 px (0.6220 em)	×1.2125	×0.825

The headline figure is ×0.816 at body size. It barely moves across the scale, because Literata's x-height is nearly flat across its optical-size axis. Every figure above was read off real rendered ink in Chromium using the Canvas text-metrics interface, not from what the font declares about itself.

Why declared metrics were not trusted. Noto Sans Bengali declares a cap height of 622 units. That is not the height of its capital H — it is the মাত্রা height. The field has been repurposed. A design system that read it to align two scripts would be aligning to the wrong thing entirely.

The floor, and the weight bump

Bangla never goes below 12 px, whatever the multiplier would otherwise say. Applying the multiplier strictly all the way down drives Bangla under the size at which its মাত্রা and its conjuncts survive.

Below 14 px Bangla gains one weight step. Measured at a device scale factor of 1: at 12 px and weight 400 the মাত্রা renders at luminance 123 on white, which reads as grey rather than black. At weight 500 it holds at 108. Body size lands at 13.1 px and caption size lower still, so both gain the step; anything at lead size or larger is exempt.

The মাত্রা stroke, measured, and why it sets a weight step.

Measure	Noto Serif Bengali
মাত্রা thickness	0.05417 em
At 16 px	0.867 device px
At 11 px	0.596 device px
Continuity across কলকাতা at 240 px	726 of 726 columns (1.000)

The thinnest মাত্রা of any text face measured, which is the price of the editorial register. It is why the weight bump exists and why the floor is not negotiable.

The floor is a **judgement built on the measurements, not itself a measurement**. The trade is between exact size parity and Bangla legibility, and legibility wins.

Line height

Bangla runs at 1.6. The collision floor — the first line-height at which a descender-heavy Bangla line stacked above a headline-heavy one separates into two bands of ink with clear white between them — was measured at 1.25. The token is the floor plus air, because ink that merely stops touching is a low bar.

Do not use the fonts' own declared line spacing. The default line box a browser computes from declared metrics is wildly inconsistent between Bangla faces — one declares 1.866 em, another 1.325. Setting `line-height: normal` would give a different rhythm per face. Set the token.

Shaping

Conjuncts passed: 16/16.

Test string	Code points	Glyphs	Missing glyph	Dotted circle
অনিন্দ্য স্টুডিও	16	11	none	none
কর্ম	4	3	none	none
বিদ্যা	6	5	none	none
স্বাস্থ্য	9	4	none	none
০১২৩৪৫৬৭৮৯	10	10	none	none

Sixteen conjuncts and five words were shaped with HarfBuzz. No dotted circles, no missing glyphs, no stray hasantas. The studio name shapes 16 code points into 11 glyphs, which is the conjunct formation working. A negative control confirmed it: a naive count gives 16 glyphs, and shaped output gives 11.

That is a test for failure, not for quality. It proves nothing is broken. Whether these conjuncts read as well-drawn Bangla to a Bangla reader is a judgement no measurement can make, and no Bangla reader has been asked.

The families this one was chosen over

Eight pairings measured out of thirty families.

Pairing	Latin	Bangla	Multiplier at 16 px	Varies with size
Core Modern	inter	notosansbengali	×0.872	yes
Technical	ibmplexsans	notosansbengali	×0.83	no
Editorial	newsreader	notoserifbengali	×0.708	yes
Rooted / Scholarly	literata	tirobangla	×0.818	yes
Rooted / Familiar	sourcesans3	hindsiliguri	×0.757	no
Civic / Systemic	publicsans	notosansbengali	×0.831	no
Contemporary	archivo	anekbangla	×0.84	no
Editorial (revised)	literata	notoserifbengali	×0.816	yes

The obvious editorial pairing was Newsreader with Noto Serif Bengali, and the measurements rejected it: at ×0.708 the Bangla column is visibly smaller and lighter than the English beside it. For a brand whose two audiences are equal, that is the one failure that is not acceptable. Literata has an x-height a full pixel taller, which lifts the multiplier to ×0.816 and holds it nearly flat across the whole scale.

The floor, per platform

The smallest text size each Apple platform allows, against the smallest step in this scale. Two of these are floors this kit clears with nothing to spare.

Platform	Default	Minimum	This kit's smallest step, 12.0032 px
iOS and iPadOS	17 pt	11 pt	clears it
macOS	13 pt	10 pt	clears it
tvOS	29 pt	23 pt	below it — this kit is not specified for tvOS
visionOS	17 pt	12 pt	clears it
watchOS	16 pt	12 pt	clears it

Source: Apple Human Interface Guidelines — Typography — developer.apple.com/design/human-interface-guidelines/typography, change log 16 December 2025. These apply to custom fonts as well as system fonts.

This kit is built for the web, where the unit is the CSS pixel and there is no platform minimum to clear — so its own floor is the one it sets: **12.0032 px** for the smallest step, and a hard **12 px** for Bangla at any size, because below that the মাত্রা stops resolving into a line. Ported to an Apple platform, that floor clears iOS, iPadOS, macOS, visionOS and watchOS, and does **not** clear tvOS, whose minimum is 23 pt. This kit is not specified for tvOS and does not claim to be.

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The word this chapter turns on

মাত্রা — the headline stroke running along the top of the letters. Every size rule in this system exists because of it. It is the height that governs how large Bangla looks, and it is the first thing to go pale when the type gets small.

The shaping test strings

The shaping test strings, with code points in and glyphs out.

String	Code points	Glyphs
অনিন্দ্য স্টুডিও	16	11
কর্ম	4	3
বিদ্যা	6	5
স্বাস্থ্য	9	4
০১২৩৪৫৬৭৮৯	10	10

These five strings and sixteen conjuncts were shaped with HarfBuzz and produced no dotted circles, no missing glyphs and no stray hasantas. The English section carries the full measurement set, the multipliers, the 12 px floor and the weight bump — all of them numbers rather than prose, identical in both languages.

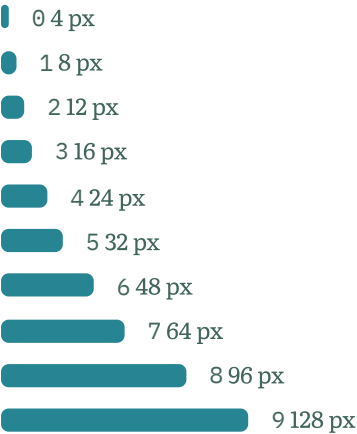
Space and shape • ফাঁক ও আকার

A 4 px scale in ten steps, four radii, four target sizes and one focus ring.

English

The space scale

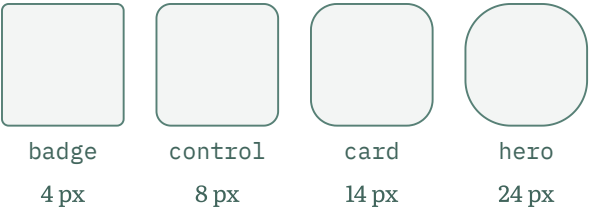
Ten steps built on 4 px. Everything in the system sits on one of them: no padding, gap or margin anywhere in the component layer is a number typed by hand.



The ten steps of the space scale.

Token	Value	At 16 px root
--as-space-0	4 px	0.25 rem
--as-space-1	8 px	0.5 rem
--as-space-2	12 px	0.75 rem
--as-space-3	16 px	1 rem
--as-space-4	24 px	1.5 rem
--as-space-5	32 px	2 rem
--as-space-6	48 px	3 rem
--as-space-7	64 px	4 rem
--as-space-8	96 px	6 rem
--as-space-9	128 px	8 rem

Four radii



The four radii, and what each is used on.

Token	Value	Where
--as-radius-badge	4 px	Badges and small pills
--as-radius-control	8 px	Buttons, inputs, selects
--as-radius-card	14 px	Cards, dialogs, panels
--as-radius-hero	24 px	The icon tile, and large surfaces

The hero radius is also the icon's corner rounding, at 24% of the icon width. The manifest records where that number comes from: read at build time from dimension.radius.hero. That token is 24 px; this is 24 units per 100 on the icon grid, which is 245.8 px of rounding on the 1024 px icon. The numeral is reused deliberately, so the icon corner belongs to the same family as every other rounded corner in the kit — it is an echo, not a unit conversion. Apple publishes no corner radius; this number is ours and is not claimed to be theirs. Apple publishes no corner radius and does not use the word squircle anywhere in current guidance.

Target sizes

A target is the area a finger or a pointer can hit. Three authorities give three different minimums, and they are not interchangeable. All four figures are tokens, so the platform decides rather than habit.

The four target sizes, each with the guidance it comes from.

Token	Size	Where the figure comes from
--as-target-min	24	WCAG 2.2 criterion 2.5.8, in CSS pixels. The floor this system is held to.
--as-target-apple-min	28	Apple's published minimum, in points, for iOS and iPadOS.
--as-target-comfortable	44	Apple's published default, in points. The size to use unless there is a reason not to.
--as-target-android-min	48	Android's minimum touch target, in density-independent pixels.

A CSS pixel is the web's device-independent unit of length. It is not the same as a physical screen pixel, an Apple point or an Android density-independent pixel, which is why the three figures above cannot be compared directly or collapsed into one number.

The focus ring

The focus ring, and the criterion behind each figure.

Token	Value	Why
<code>--as-focus-ring-width</code>	3 px	WCAG 2.2 criterion 2.4.13 asks for a perimeter at least 2 CSS pixels thick. 3 px clears it with room for a rounded corner.
<code>--as-focus-ring-offset</code>	2 px	Separates the ring from the control's own border so the two do not read as one thick edge.

The ring is drawn on `:focus`, not `:focus-visible`. `:focus-visible` is a browser heuristic, and a heuristic can decide not to draw the ring. Showing it once too often is a smaller failure than losing it once.

Apple publishes **no numeric focus-indicator specification at all**, and its focus guidance page predates its current material system entirely. There was nothing to inherit here, so this system specifies its own numbers and meets WCAG's instead.

The gap between two controls

Adjacent controls are separated by at least `--as-space-2`, which is 12 px. Where controls have no visible edge of their own — a row of quiet buttons, an icon-only toolbar — the gap goes to `--as-space-4`, 24 px. That is the rule; the component layer already obeys it, and now says so.

Apple treats spacing between controls as as important as size, suggesting roughly 12 pt of padding around bevelled elements and roughly 24 pt around unbevelled ones. Source: Apple Human Interface Guidelines — Accessibility — developer.apple.com/design/human-interface-guidelines/accessibility, change log 9 June 2025. The two figures coincide with two steps this scale already had, which is why no token was added for them. A size without a gap is half a specification: two 24 px targets touching are harder to hit accurately than one, however well each measures.

Bangla · বাংলা

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This chapter is ten numbers, four numbers, four numbers and two numbers. None of them changes between languages, and the English section carries all of them with the reason for each.

Every step of every scale in this chapter, in one place. All values in px, read from the token file.

Scale	Steps
Space	4, 8, 12, 16, 24, 32, 48, 64, 96, 128
Radii	4, 8, 14, 24
Targets	24, 28, 44, 48
Focus ring	3 wide, 2 offset

Components • উপাদান

Thirty cards in three groups, and the three rules the component layer is built to keep.

English

30 cards — 6 foundations, 16 components, 8 patterns. Each is a single self-contained HTML file with the tokens, the component layer and the three fonts inlined. A card opens from a file path with no network at all.

The three rules the layer enforces

1. **No literal colour.** Every colour in the component layer is a `var(--as-...)` from the token file. The build scans for `hex()`, `rgb()`, `hsl()`, `lab()`, `oklch()` and the CSS named colours, and refuses to build if it finds one. Two keywords are allowed and nothing else: `currentColor`, which is a reference rather than a value, and `transparent`, which is the only way CSS lets you express the absence of a colour.
2. **Focus is always visible.** Every focusable control gets a ring from `:focus`. No rule in the layer sets `outline: none`.
3. **Nothing relies on colour alone.** Every state carries a word and a glyph as well as a colour — a danger badge says Failed and shows a cross, a selected tab is bold and underlined, the current navigation item has a bar.

CSS cannot enforce rule 3, and this book will say so. A stylesheet has no way to know whether the markup it is styling carries a word next to the colour. A danger modifier will happily colour an empty element red. Rules 1 and 2 are machine-checked. Rule 3 lives in the markup and in review, and it is a promise rather than a guarantee.

Glyphs are drawn, not typed

Every glyph in the system is an inline SVG in `currentColor`. Literata has no tick, no cross and no warning triangle, so a glyph typed as a character would fall back silently to whatever font the reader's machine happens to have. Drawing them keeps the shape, the weight and the colour under the system's control, and `currentColor` means each glyph inherits both the theme and the forced-colours palette.

Foundations — 6

The foundations cards, read from 08_components/_cards.json.

Card	Bangla	What it shows	Design canvas
Colour	রং	Every colour role across four themes, each with the contrast ratio it was measured at and the criterion it was measured against, over the seven surfaces they are measured against.	1280 × 2400
Typography	হরফ	One scale of a perfect fourth, two scripts, a measured multiplier for Bangla and a floor it never goes below.	1280 × 1900
Space and shape	ফাঁক ও আকার	A 4 px scale in ten steps, and four radii. Everything in the system sits on one of them.	1280 × 1500
Motion	গতি	Two durations and three easing curves. Things that move may overshoot; things that only change colour never do.	1280 × 1500
The marks	চিহ্ন	The mark in two weights, drawn in currentColor so it takes whatever theme it lands in.	1280 × 1400
Accessibility	অভিগম্যতা	Target sizes with the guidance each one comes from, the anatomy of the focus ring, and what happens in forced colours — the mode where the operating system replaces every colour with its own.	1280 × 1700

Components — 16

The components cards, read from 08_components/_cards.json.

Card	Bangla	What it shows	Design canvas
Button	বটন	Four kinds, two sizes and an icon-only form, each with a label that says what will happen.	1280 × 1500
Input	ইনপুট	A label, an optional hint, and an error that says what happened and then what to do next.	1280 × 1500
Select	বাছাই তালিকা	A native select with a drawn arrow, so the arrow follows the theme instead of the operating system.	1280 × 1300
Checkbox and radio	চেকবক্স ও রেডিও	Native controls at 24 px, wrapped in a label so the words are part of the target.	1280 × 1500
Textarea	লেখার ঘর	You can drag it taller but never wider, so the line length stays comfortable to read.	1280 × 1300
Badge	ব্যাজ	Five meanings, each carrying a glyph and a word so the colour is the third signal and never the only one.	1280 × 1200
Card	কার্ড	A surface a step brighter than the page, with a shadow in the light theme and none in the dark ones.	1280 × 1200
Alert	বার্তা	Four kinds. Each says what happened, then what happens next, and never blames the reader.	1280 × 1600
Dialog	ডায়ালগ	A real dialog element over a dimmed backdrop, with the destructive action named rather than called OK.	1280 × 1200
Table	টেবিল	Row headers, a caption saying what the numbers are, and a sideways scroll when the table is wider than the space.	1280 × 1200
Tabs	ট্যাব	The selected tab is bold, underlined and marked with aria-selected. Three signals, one of which is a colour.	1280 × 1200
Nav	নেভিগেশন	Vertical and horizontal. The current item carries a bar, a heavier weight and aria-current.	1280 × 1200
Breadcrumb	পথরেখা	The last item is not a link, because you are already on it.	1280 × 1000
Toast	টোস্ট	A short message with a dismiss button that has a name of its own, not only a cross.	1280 × 1200
Empty state	যখন কিছু নেই	Says what is missing, and then exactly what to do about it.	1280 × 1300
Code block	কোড ব্লক	Aninda Mono, a horizontal scroll rather than a wrap, and a copy button that says what it copies.	1280 × 1100

Patterns — 8

The patterns cards, read from 08_components/_cards.json.

Card	Bangla	What it shows	Design canvas
Sign in	প্রবেশ	One card, two fields, and an option for someone who has no password.	1280 × 1500
Settings	সেটিংস	Grouped in fieldsets, with the destructive action kept apart and named.	1280 × 1900
Dashboard	ড্যাশবোর্ড	Four figures, one table, and a note saying where the numbers came from.	1280 × 1900
Docs page	নির্দেশিকার পাতা	Breadcrumb, page navigation and prose held to a readable line length.	1280 × 1900
Landing	প্রথম পাতা	A claim, the reason to believe it, and two ways forward.	1280 × 1900
Pricing	দাম	Three plans, with the recommended one marked by a badge and a word.	1280 × 1600
Not found	পাতা পাওয়া যায়নি	Says the page is missing, then offers the pages most people were looking for.	1280 × 1300
Form with validation	যাচাইসহ ফর্ম	A summary at the top, an error under each field, and nothing lost.	1280 × 1700

The fonts each card carries

Every font artefact this kit redistributes: 3 subsets and 1 whole face, each with its licence and its size as written to disk. The whole face is the desktop file, listed here because a reader is more likely to install or pass on that one than any subset.

Family	Licence	Reserved Font Name	Form	Size on disk	Licence text
Literata	SIL OFL 1.1	No	Subset	82 kB	fonts/literata-OFL.txt
Noto Serif Bengali	SIL OFL 1.1	No	Subset	111 kB	fonts/notoserifbengali-OFL.txt
Aninda Mono	SIL OFL 1.1	Yes — renamed	Subset	10 kB	fonts/anindamono-OFL.txt
Aninda Mono (desktop)	SIL OFL 1.1	Yes — renamed	Whole face	136 kB	fonts/anindamono-OFL.txt

How the cards are checked

Every card is opened in a real Chromium at 360, 768 and 1280 CSS pixels and in all four themes, and then measured. A static reading of the CSS can tell you a rule exists; it cannot tell you whether the rule reached the pixel.

- Contrast is read off the composited effective background, walking ancestors and blending partly transparent layers. Reading an element's own background colour returns a fully transparent value for nearly every element in a real page and proves nothing.
- Interaction states are driven by a real pointer — move, down, up — and the pressed style is compared against the hovered one. Two identical readings are what a dead rule looks like from outside the browser.
- The focus indicator is measured from pixels: the element is captured unfocused and focused, the two buffers are differenced, and the changed pixels are checked for a ring at least 2 CSS pixels thick at 3:1.
- Forced colours runs a liveness probe first. If the emulation is inert the run fails as not-equipped rather than passing silently. A check that cannot fail is not a check.

Bangla · বাংলা

The title of this chapter is verified. Its body prose is not written.

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The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

The cards that do have a verified Bangla name

The cards that do have a verified Bangla name.

Bangla	English	Group
রং	Colour	Foundations
হরফ	Typography	Foundations
ফাঁক ও আকার	Space and shape	Foundations
গতি	Motion	Foundations
চিহ্ন	The marks	Foundations
অভিগম্যতা	Accessibility	Foundations
বাটন	Button	Components
ইনপুট	Input	Components
বাছাই তালিকা	Select	Components
চেকবক্স ও রেডিও	Checkbox and radio	Components
লেখার ঘর	Textarea	Components
ব্যাজ	Badge	Components
কার্ড	Card	Components
বার্তা	Alert	Components
ডায়ালগ	Dialog	Components
টেবিল	Table	Components
ট্যাব	Tabs	Components
নেভিগেশন	Nav	Components
পথরেখা	Breadcrumb	Components
টোস্ট	Toast	Components
যখন কিছু নেই	Empty state	Components
কোড ব্লক	Code block	Components
প্রবেশ	Sign in	Patterns
সেটিংস	Settings	Patterns
ড্যাশবোর্ড	Dashboard	Patterns
নির্দেশিকার পাতা	Docs page	Patterns
প্রথম পাতা	Landing	Patterns
দাম	Pricing	Patterns
পাতা পাওয়া যায়নি	Not found	Patterns
যাচাইসহ ফর্ম	Form with validation	Patterns

30 of the 30 cards carry a verified Bangla name and 30 carry a verified Bangla subtitle. Every string here comes from the approved register; nothing on a card was translated for the card.

Motion • গতি

Two durations, three curves, and why Material's spring system was read and not adopted.

English

120 ms for a colour change. 220 ms for something arriving or leaving. Nothing in this system goes over 300 ms.

The two durations, and what each is for.

Token	Value	What it is for
--as-duration-colour	120 ms	Anything that changes in place: a hover tint, a border, a background, a text colour.
--as-duration-move	220 ms	Anything that arrives or leaves: a dialog, a toast, a panel, a menu.

Three curves

A cubic Bézier curve is four numbers describing how a change accelerates and slows. These three are the whole set.

The three easing curves, drawn from their own control points.

Token	Curve	Shape	Where
--as-ease-standard	cubic-bezier(0.2, 0, 0, 1)		Anything that stays on screen throughout — a colour, a size, a position.
--as-ease-enter	cubic-bezier(0.05, 0.7, 0.1, 1)		Something arriving. Fast at the start, settling at the end.
--as-ease-exit	cubic-bezier(0.3, 0, 0.8, 0.15)		Something leaving. Slow to commit, then quick to go.

Material's spring system was read, and not adopted

Material 3 publishes a spring-based motion system: two schemes, each with six specifications on a grid of three speeds by two kinds of motion. The expressive scheme's spatial values are damping 0.8 with stiffness 380 at default, 0.6 with 800 fast, and 0.8 with 200 slow; its effects values are damping 1.0 with stiffness 1600, 3800 and 800. Those figures were checked on 14 August 2026.

This system does not adopt them, for three reasons. A spring needs a physics integrator at run time, and this kit is a stylesheet — it has no run time of its own. A spring's duration is an outcome rather than a setting, which makes it far harder to hold to a stated ceiling. And Material did not replace its own duration and easing tokens when it added springs: 16 durations and 10 easing curves remain published. A kit built on durations and cubic Bézier curves is not out of date relative to Material. It is using the other half of a system that publishes both halves.

The principle underneath their numbers does transfer, and this system takes it. Every effects damping in Material's scheme is exactly 1.0 — critically damped, meaning it never overshoots. Every spatial damping is below 1.0 — underdamped, meaning it overshoots and settles. So: **things that move may overshoot; things that only change colour never do.** That is why `--as-duration-colour` and `--as-duration-move` are two tokens and not one, and why the enter and exit curves are only ever applied to things that arrive or leave.

Apple publishes no durations at all

Apple's motion guidance is principled rather than numeric: add motion purposefully, keep it brief, show restraint on frequent interactions, let people cancel it. The only two figures anywhere in it are that games should run at 30 to 60 frames per second, and that visionOS should avoid sustained oscillation at around 0.2 Hz. Neither applies to a studio identity system. **Every duration and curve in this kit is this kit's own, and presenting any of them as Apple-aligned would be false.**

Reduced motion

When the reader has asked their system for reduced motion, both durations collapse to 1 ms at the root. Nothing is left half-animated and nothing needs a second rule further down the tree.

```
@media (prefers-reduced-motion: reduce) {
  :root {
    --as-duration-colour: 1ms;
    --as-duration-move: 1ms;
  }
}
```

The substitution rule, inherited from Apple's own reduced-motion techniques, is to replace a movement with a **fade** — never a spatial move and never a blur, and never an animation into or out of a blur.

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The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

The motion tokens.

Token	Value
--as-duration-colour	120 ms
--as-duration-move	220 ms
--as-ease-standard	cubic-bezier(0.2, 0, 0, 1)
--as-ease-enter	cubic-bezier(0.05, 0.7, 0.1, 1)
--as-ease-exit	cubic-bezier(0.3, 0, 0.8, 0.15)

Five tokens, and one rule that holds them together: things that move may overshoot, things that only change colour never do. The English section sets out where that rule came from and why Material's spring system was read and not adopted.

Voice • কণ্ঠস্বর

Plain international English, British spelling, and the six words that are banned outright.

English

Plain international English, British spelling. That is the rule in one line, and it applies to every English word the studio ships: interface text, error messages, documentation, this book, README files, the website, proposals and invoices.

Why those two things together

They sound as though they pull in opposite directions. They do not, because they answer two different questions.

- **Vocabulary and sentence construction — international.** Written so a reader whose first language is not English understands it on one pass. That is the part that decides whether the writing works.
- **Spelling — British.** Bangladesh's education system and press use British spelling, so *colour*, *organise* and *centre* are what a Bangladeshi reader already expects. A spelling convention has to be picked, and picking the one both audiences already read is the sensible answer.

So *colour*, never *color*. But also never *whilst*, *amongst*, *sort out* or *quid*.

The standard being followed

ISO 24495-1:2023 — Plain language, Part 1: Governing principles and guidelines. It is the international standard for plain language. In its own terms, readers get what they need if the writing is **relevant**, **findable**, **understandable** and **usable**.

Sentences

- One idea per sentence. Around 15 to 20 words on average. Break anything over 25.
- Active voice by default. *The build failed*, not *a failure was encountered*.
- Say who does what. A sentence with no subject usually needs one.
- Main point first, then the detail. A reader who stops after one sentence should still have the answer.

Words

- The shortest word that is still precise. use not utilise, about not approximately, start not commence, enough not sufficient.
- **Precision beats plainness when they conflict.** If a technical term is the correct one, keep it and explain it in one sentence the first time it appears. *Contrast ratio* stays; it gets a definition.
- **No idioms and no phrasal verbs where a plain verb exists.** cancel not call off; continue not carry on. Idioms are the single largest cause of misreading for a reader whose first language is not English.
- **No metaphors from culture or sport.** No *ballpark*, no *touch base*, no *out of the box*.
- **No Latin abbreviations.** Write the words out.
- **Say the number and its unit.** 10 MB, 220 ms, 4.5:1. Never a *large file* where a figure is known.
- One name for one thing, everywhere. If it is the *mark*, it is never also the *logo* on the next page.

Banned outright

simply · just · easy · obviously · of course · clearly

Every one of them tells a reader who is stuck that the problem is them. There are no exceptions, and the build checks for them: if one of those words reaches this book, the build fails and writes nothing. The list above is the only place in the book where they appear, and it is marked in the markup so the check skips it.

Banned: **e.g.**, **i.e.**, **etc.**. Write *for example*, *that is*, and *so on*.

No exclamation marks. Warmth comes from the words, not from the punctuation.

Tone, in four words and what each one means at the keyboard

Cooperative — write as if sitting beside the reader, not across a desk. Offer the next step rather than stating the position. Use *you* and *I* in the same sentence when it is a joint problem. When there is a choice, give the recommendation and the reason, then leave the decision with them.

Friendly — warm, not chummy. Contractions where they fall naturally. Thanks for a real thing, not as a reflex. No *Hi there*, no *Awesome*, no *Happy to help*, no emoji.

Approachable — never make the reader feel they should already have known. Explain the term the first time. Say plainly when something is genuinely difficult; it is a relief to read. Invite the question.

Professional — **first person singular**. It is one person, so *I*, never *we*. Using *we* to sound larger is the first small dishonesty a studio tells about itself. Be accurate before being warm. Name a limit where it would otherwise surprise someone — as information, not as a confession, and never wrapped in an apology.

When two of them pull against each other: accurate first, then clear, then warm. A sentence is never made friendlier at the cost of being right.

Interface text

- **Buttons are a verb with its object.** *Save the entry*, not *Submit*, not *OK*. A button whose label does not say what will happen is a trap.
- **Errors say what happened, then what happens next.** Never *Error 500*, and never blame the reader.
- **Empty states say what to do.**
- Sentence case for everything — headings, buttons, labels.
- Never rely on colour alone. Every state carries a word and a glyph as well.

The four interface strings, in both languages

These are the reference strings the whole system is written against. The Bangla is not a translation of the English; each was written in its own language and then checked against the Bangla Academy rules.

The verified button and message strings, English first.

English	Bangla
Save the entry	লেখাটি সংরক্ষণ করুন
Cancel	বাতিল করুন
Delete the file	ফাইলটি মুছে ফেলুন
Try again	আবার চেষ্টা করুন
Copy the code	কোডটি কপি করুন
Couldn't save. Your work is still here — try again in a moment.	সংরক্ষণ করা যায়নি। আপনার লেখা এখনো আছে — একটু পরে আবার চেষ্টা করুন।
That file is too large. 10 MB at most.	ফাইলটি অনেক বড়ো। সর্বোচ্চ ১০ মেগাবাইট।
Nothing here yet. Add your first entry to begin.	এখনো কিছু নেই। শুরু করতে প্রথম লেখাটি যোগ করুন।
Saved	সংরক্ষিত হয়েছে

What this does not mean

- **Not dumbed down.** Plain language is about being understood, not about having less to say.
- **Not shorter at any cost.** A sentence that saves three words and costs a re-read is a bad trade.
- **Not the same as the Bangla.** The Bangla is written as Bangla. Where the two must agree exactly — a number, a licence name, a file name — they agree. Everywhere else they are written independently.

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The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

The register

চলিত throughout. No সাধু forms anywhere in the product. That is the Academy's own register: rule 2.11 replaces সাধু forms with চলিত ones outright, and the whole of section 5 gives its ten verb paradigms exclusively in চলিত.

The voice sample

আমি ছোটো, যত্নে গড়া সফটওয়্যার বানাই। কোনো কিছুইর সীমা থাকলে সেটা এখানেই লেখা থাকবে — লুকিয়ে রাখা হবে না।

Reviewed word by word. সফটওয়্যার is the dictionary's exact headword spelling, free of the hasanta under rule 2.10. কোনো is correctly the indefinite form. বানাই is plain চলিত — far better than তরি করি, which would make a one-person studio sound like a ministry. এখানেই carries the নিশ্চয়্যার্থক ই attached in full under rule 3.5. The em dash is a genuine Bangla যতিচিহ্ন, not an English import.

The verified interface strings

Every Bangla string the system ships. Each was reviewed against the Bangla Academy's 2012 rules and its dictionary, string by string.

Buttons.

Bangla	English
লেখাটি সংরক্ষণ করুন	Save the entry
বাতিল করুন	Cancel
ফাইলটি মুছে ফেলুন	Delete the file
আবার চেষ্টা করুন	Try again
কোডটি কপি করুন	Copy the code

Messages.

Bangla	English
সংরক্ষণ করা যায়নি। আপনার লেখা এখনো আছে — একটু পরে আবার চেষ্টা করুন।	Couldn't save. Your work is still here — try again in a moment.
ফাইলটি অনেক বড়ো। সর্বোচ্চ ১০ মেগাবাইট।	That file is too large. 10 MB at most.
এখনো কিছু নেই। শুরু করতে প্রথম লেখাটি যোগ করুন।	Nothing here yet. Add your first entry to begin.
সংরক্ষিত হয়েছে	Saved

Two rulings worth keeping in the head

এখনো, **not** এখনও. The Academy dictionary makes এখনও a bare cross-reference — "দ্র এখনো" — to the full entry at এখনো.

ছোটো, **not** ছোট, **and** বড়ো, **not** বড়. Rule 2.3 names ছোটো in its own example list alongside কালো, খাটো and ভালো, and both ছোটো and বড়ো are dictionary headwords while ছোট and বড় have no entry at all.

Writing in Bangla • বাংলায় লেখা

Standard Bangla by the Bangla Academy's own rules, and the policy of never inventing any.

English

The Bangla in this system follows প্রমিত বাংলা — standard Bangla — as set out by the Bangla Academy.

Which document is current

The governing document is বাংলা একাডেমি প্রমিত বাংলা বানানের নিয়ম: first edition 1992, revised edition September 2012, first reprint of the revised edition January 2015. Searches on 14 August 2026 found no later edition. The 2012 text, reprinted in 2015, remains the standard.

The second source used throughout is the Academy's own dictionary, আধুনিক বাংলা অভিধান. Where the rules are permissive, the dictionary's choice of headword is the tie-breaker, because it is the Academy applying its own rules.

The rule this whole system runs on

Only verified Bangla is shipped. No Bangla is invented.

Every Bangla string in this book, in the component library and in the token files was reviewed string by string against those two sources. Where a verified string exists, it is used. Where one does not, the English is left in place and the gap is named. That is why parts of the Bangla sections of this book say so plainly instead of quietly filling the space.

This is not modesty. Inventing Bangla and shipping it looks the same as having written it well, right up until a Bangla reader meets it.

The ten house rules

1. এখনো, কখনো, তখনো with ও-কার — but আরও with the full ও. The asymmetry is the Academy's own: the dictionary makes এখনও a bare cross-reference to এখনো, while it calls আরও the সংগততর — more consistent — spelling of আরো.
2. কোনো means *any* or *some*. কোন means *which*. Never interchange them.
3. রং standing alone or at the end of a word. রঙ only when a vowel follows — রঙিন, রঙের.
4. ছোটো, বড়ো, ভালো, কালো, মতো, হলো, হতো — ও-কার, under rule 2.3.
5. No গ in any non-Sanskrit word: ধরন, রানি, সোনা, বরনা, ঠান্ডা.
6. Avoid the hasanta where it is not needed: সফটওয়্যার, not সফ্টওয়্যার.

7. Established loanwords are standard Bangla: কপি, ফাইল, সফটওয়্যার, মেগাবাইট. Do not "correct" them into অনুলিপি or নথি — that would be less standard, not more.
8. চলিত **throughout**. No সাধু forms anywhere in the product.
9. **Bengali numerals** in prose and in measurements. Western digits only where a string has to be copied literally — a version number, a code sample, a file extension.
10. The দাঁড়ি ends a statement. The question mark and the exclamation mark are the ordinary Western characters. No space before the দাঁড়ি, one space after, and never let it begin a line.

Rule 9 applies to Bangla prose, and this book's prose is English, so every rule number quoted here uses Western digits. There is a practical reason as well as a consistency one: the Bengali digit for four is drawn as an 8, so a Bengali numeral inside an English sentence reads as the wrong number to a reader who does not know the script.

Punctuation

The Academy's 2012 rules contain no punctuation section. The rules below come from the W3C's Bengali script documentation and from the Bengali Wikipedia article on যতিচিহ্ন.

Statement । U+0964 · question ? U+003F · exclamation ! U+0021. Phrase level: , ; : — the ordinary Western marks. There is no Bengali-specific question or exclamation mark.

The W3C's *Bengali Layout Requirements* records two approaches to spacing around the দাঁড়ি and names the primary one: no space character before it, letting the font's own advance width open a small gap, then a single space after. It also records a line-breaking constraint — never move a দাঁড়ি to the beginning of a new line.

The em dash is a genuine Bangla যতিচিহ্ন, not an English import. It is used in this system's Bangla error message and in the voice sample, doing exactly the job it is meant for. It is half as long again as the hyphen, which is a separate mark with a separate job.

Latin words inside Bangla sentences

Normal, and sanctioned by the Academy's own practice — its dictionary embeds Latin script inside Bangla entries constantly, as glosses and as technical terms.

1. Where the dictionary has a Bangla form, prefer it in prose. Write মেগাবাইট inside a Bangla sentence, not "MB".
2. Keep Latin for product names that must stay recognisable and searchable — GitHub, Figma, CSS.
3. Keep Latin for anything copied literally.
4. Do not mix scripts inside one word. GitHub-এ — Latin name, Bangla case-ending, hyphenated — is the established Bangladeshi convention. It is marked here as a convention, because no formal style-guide citation for it was found.

Setting Bangla type

The type chapter carries the measured numbers. The two that matter most when writing rather than designing:

- Bangla never goes below **12 px**, whatever the size relationship would otherwise say.
- Below **14 px** Bangla gains one weight step, because the মাত্রা — the headline stroke running along the top of the letters — goes pale before the letters do.

Where this system's Bangla comes from

The Bangla this book rests on. The first three rows are counted from the recommended-strings table of 06_type/BANGLA-STANDARD.md; the fourth is the register in 06_type/bangla-strings.json merged over it, which is where most of what you can see comes from.

Measure	Count
Reviewed strings	31
Changed by the review	6
Unchanged	25
Verified strings available to this book	125
Component cards with no verified Bangla name	0 of 30
Component cards with no verified Bangla subtitle	0 of 30

What is explicitly unverified

- Any Bangla Academy edition later than 2012. None was found as at 14 August 2026.
- Any Academy rule on ৭ or on চন্দ্রবিন্দু. No such rule exists in the 2012 document; both are lexical, so follow the dictionary word by word.
- Any Academy or published style-guide rule on Bengali against Western numerals. None was found. Rule 9 above rests on the Academy's own printing practice plus one editorial judgement about strings that must survive copy and paste.
- The GitHub-এ hyphenation convention. Widespread practice, no formal citation.
- তখনো as a dictionary headword. Neither তখনো nor তখনও has an entry; rule 1 infers it from এখনো and কখনো, and says so.

And the largest one: no second Bangla reader has reviewed any of this. The sources were read carefully and quoted with page numbers. That is a reading, not a review. Chapter 14 repeats it, because it belongs in both places.

The title of this chapter is verified. Its body prose is not written.

Only Bangla that has been checked against the Bangla Academy's 2012 spelling rules and its own dictionary is shipped in this system. No Bangla is written for this book. Where a verified string does not exist, the English stays and the gap is named here and in chapter 14.

The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

There is a second reason this chapter has no Bangla body, beyond the general one. A chapter that tells you how to write standard Bangla should not be the place where unreviewed Bangla is introduced. The English section carries the ten house rules, the punctuation rules and the full list of what is unverified.

The words the rules turn on

Each of these is quoted from the Bangla Academy's 2012 rules or from আধুনিক বাংলা অভিধান. None of them was written for this book.

The spellings this book uses, the ones it does not, and the ruling behind each.

Use this	Not this	Why
এখনো	এখনও	এখনও is only a cross-reference to এখনো in the dictionary
আরও	আরো	The dictionary calls আরও the সংগততর — more consistent — spelling
কোনো	কোন	কোনো is <i>any</i> or <i>some</i> ; কোন is <i>which</i>
রং	রঙ	Rule 2.4 names রং; রঙ has no dictionary entry
রঙিন, রঙের	—	রঙ appears only when a vowel follows it
ছোটো, বড়ো	ছোট, বড়	Rule 2.3; both are dictionary headwords, neither short form has an entry
ধরন, রানি, সোনা, বরনা	—	Rule 2.7: no ণ in any non-Sanskrit word
সফটওয়্যার	সফ্টওয়্যার	Rule 2.10: avoid the hasanta where it is not needed
কপি, ফাইল, মেগাবাইট	অনুলিপি, নথি	Rule 2.6: established loanwords are standard Bangla
পদ্ধতি	ব্যবস্থা	ব্যবস্থা reads administrative in Bangladeshi usage

The chapter title itself

The Bangla title of chapter 14 is যা এই পদ্ধতি করে না, and it is one of the six strings the review changed. The draft used ব্যবস্থা, which is correct in the abstract but lives in official notices — ব্যবস্থা নেওয়া, আইনগত ব্যবস্থা — and imports exactly the institutional tone that chapter is trying to disown. পদ্ধতি is a dictionary headword and the ordinary Bangla for a system in technical writing, as in সংখ্যা পদ্ধতি.

Applying it • কাজে লাগানো

How to install it, theme it, check it, and rebuild this book.

English

Everything in this chapter is about the design system half — the part under Apache-2.0, which you may use without asking. The identity half is covered in chapter 13.

The shortest possible start

```
<link rel="stylesheet" href="tokens.css">
<link rel="stylesheet" href="components.css">
```

`tokens.css` defines every custom property. `components.css` is the component layer, and it contains no literal colour at all — every colour in it is a `var(--as-...)` from the token file. That is machine-checked at build time, so a hard-coded colour cannot reach it.

Then write markup with the `.as-` classes, inside an element carrying `.as-root`.

Or install it

Not published yet. On 2026-08-18 I checked the npm registry and the Python Package Index, and neither holds `aninda-studio-tokens`. The two commands below are what will work once they are published. Until then, use the checkout: the packages are built, and the files they carry are in `12_packages/`.

```
npm install aninda-studio-tokens
```

```
pip install aninda-studio-tokens
```

Both packages carry the same token data: the DTCG JSON, the combined CSS, and a separate stylesheet per theme.

Themes

There are four: light, dark, high contrast light, high contrast dark. They are not four skins over one palette. Each was generated against its own contrast target, which is why the high contrast pair is a

different set of colours rather than the same colours pushed further apart.

The token file arranges them in five layers, and the order matters:

1. **The default.** Light values on `:root`, unscoped.
2. **The reader's system setting**, under `prefers-color-scheme` and `prefers-contrast`, applied only where nobody has chosen explicitly. The `:not([data-theme])` in those rules is what makes an explicit choice further down the tree possible at all.
3. **An explicit choice**, `[data-theme="..."]` on any element. These come last so they win, and because they can sit on any element, a dark panel can live inside a light page.
4. **Forced colours.** The operating system supplies the palette and every brand value gives way. A hex that survives this mode has defeated it.
5. **Reduced motion**, honoured at the root.

```
<html data-theme="dark">
<div data-theme="hc-light"> ... a high-contrast island ... </div>
```

Leave `data-theme` off entirely to follow the reader's own system setting. That is the right default for most pages.

Bangla

The token file already carries the rule:

```
:lang(bn), [lang="bn"] {
  font-family: var(--as-font-bangla);
  line-height: var(--as-bangla-line-height);
  font-size: clamp(var(--as-text-bangla-min),
                   calc(1em * var(--as-bangla-scale-body)), 100em);
}
```

The `clamp()` applies the measured size multiplier and refuses to go below the 12 px floor, so the rule and its exception live in one declaration rather than relying on anyone remembering the exception.

Mark Bangla with `lang="bn"` and the rest follows. Add `.as-bn-large` on anything at lead size or larger, which exempts it from the small-size weight bump.

Accessibility, in the order it usually goes wrong

- **Focus.** The component layer draws the ring on `:focus`, not `:focus-visible`. `:focus-visible` is a browser heuristic, and a heuristic can decide not to draw the ring. Showing the ring once too often is a smaller failure than losing it once. No rule in the layer sets `outline: none`.
- **Colour alone is never the signal.** Every state carries a word and a glyph as well as a colour. CSS cannot enforce this — a stylesheet has no way to know whether the markup it is styling carries a

word next to the colour — so it lives in the markup and in review, and it is named here as a promise rather than as a guarantee.

- **Targets.** Three different minimums exist and they are not the same number: WCAG 2.2 says 24 CSS px, Apple says 28 pt minimum with 44 pt comfortable, and Android says 48 dp. All four figures are tokens. Pick by platform, not by habit.
- **Glyphs are drawn, not typed.** Literata has no tick, no cross and no warning triangle, so a glyph typed as a character would fall back silently to whatever font the reader's machine happens to have. Every glyph in the system is an inline SVG in `currentColor`, which means it inherits both the theme and the forced-colours palette.

Checking your work

The component library ships a harness that measures rather than asserts. It opens every card in a real Chromium at three widths and in all four themes, and then reads the pixels.

```
export PLAYWRIGHT_BROWSERS_PATH=./00_sandbox/browsers
./venv/bin/python 08_components/check.py
```

It reads contrast off the composited effective background, walking ancestors and blending partly transparent layers, because reading an element's own `background-color` returns a transparent value for nearly every element in a real page and proves nothing. It drives interaction states with a real pointer. It measures the focus ring from differenced pixel buffers. It runs a liveness probe on the forced-colours emulation first, and fails as not-equipped rather than passing silently if the emulation is inert — a check that cannot fail is not a check.

It prints what it could not check at the end. That list is part of the result.

Rebuilding this book

From the repository root, as with every command in this book:

```
export PLAYWRIGHT_BROWSERS_PATH=./00_sandbox/browsers
./venv/bin/python 09_guidebook/build.py
./venv/bin/python 09_guidebook/scripts/pdf.py
```

`build.py --check` regenerates every byte in memory and compares it against what is on disk, writes nothing, and exits non-zero on the first difference. That is the drift guard. If a token moves and this book is not rebuilt, the check fails.

The two files this book ships as

What this build writes, and what each file is for.

File	What it is
Aninda-Studio-Guidebook.html	The whole kit inline. Every file in the kit is a download link whose href is its own data URI. Opens from a file path with no network.
Aninda-Studio-Guidebook-print.html	The same book with the download data URIs stripped and their labels kept as text. Adds A4 page geometry and page-break rules.
Aninda-Studio-Guidebook.pdf	A4, printed from the second file by Playwright. No other tool is involved: no ghostscript, no qpdf.

Why two HTML files, and what was measured. The interactive file carries 74 kit files — about 11.0 MB on disk, about 14.6 MB once base64-encoded. A paper page cannot be clicked, so those payloads buy a printed reader nothing.

They also cost a great deal. Printing the print build gives a PDF of 1.9 MB, read from the file in this repository. Printing the interactive build gave one of about 14.2 MB when that was measured, once, while deciding this split; that PDF is not shipped, because Chromium carries every data URI into the PDF as a link target — for links nobody on paper can follow. The interactive build also has no A4 page geometry and no page-break rules, so its layout breaks in the wrong places throughout.

One correction, since this split is usually justified with a stronger claim. Chromium's PDF pipeline is known to emit blank pages once the inlined base64 gets large, at somewhere around 24 MB. *That was not reproduced here.* At the present size the interactive build printed, with no blank page. The split rests on the two costs above, not on a failure this build has seen. Run `scripts/pdf.py --probe-interactive` to re-measure it after the kit grows.

Fonts and images stay in both builds. Without the embedded fonts the Bangla prints in a fallback face, which is the one thing this book must not do.

What is in the kit

What the kit contains, counted from the repository each time this book is built.

Measure	Value
Files in the kit	74
Total size on disk	11.0 MB
Themes	4 — light, dark, high contrast light, high contrast dark
Component cards	30 — 6 foundations, 16 components, 8 patterns
Identity files	10
Typefaces	3, each SIL OFL 1.1

The full list, with a download link for each, is at the end of this book.

The title of this chapter is verified. Its body prose is not written.

Only Bangla that has been checked against the Bangla Academy's 2012 spelling rules and its own dictionary is shipped in this system. No Bangla is written for this book. Where a verified string does not exist, the English stays and the gap is named here and in chapter 14.

The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

Everything in the English section is code, file names, package names and command lines. None of it changes between languages, and none of it should; rule 3 of the house guidance on Latin words inside Bangla says to keep Latin for anything copied literally.

The one rule that is about Bangla

Mark Bangla with lang="bn" and the token file does the rest.

```
:lang(bn), [lang="bn"] {
  font-family: var(--as-font-bangla);
  line-height: var(--as-bangla-line-height);
  font-size: clamp(var(--as-text-bangla-min),
    calc(1em * var(--as-bangla-scale-body)), 100em);
}
```

The clamp() applies the measured multiplier and refuses to go below the 12 px floor. Add .as-bn-large to anything at lead size or larger to exempt it from the small-size weight bump.

What this build writes, and what each file is for.

File	What it is
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Fonts and images stay in both builds. Without the embedded fonts the Bangla prints in a fallback face, which is the one thing this book must not do.

Licence and trademarks • লাইসেন্স ও ট্রেডমার্ক

Three licences, one thing with no licence at all, and where the boundary falls.

English

Three licences and one thing that has no licence at all.

Each part of the kit, its licence terms, and the SPDX identifier to write in a file header.

Part	Terms	SPDX identifier
The design system — tokens, stylesheets, build scripts, component cards	Apache License 2.0	Apache-2.0
The writing — the chapters of this book	PolyForm Noncommercial 1.0.0	PolyForm- Noncommercial-1.0.0
The typefaces	SIL Open Font Licence 1.1	OFL-1.1
The name, the mark, the wordmark, the tile and every lockup of them	Not licensed	—

The system: Apache-2.0

The tokens, the stylesheets, the build scripts and the thirty component cards are under the Apache License, Version 2.0. It is an OSI-approved permissive open-source licence. Use it, fork it, build a product on it, sell that product.

Keep the copyright notice and the NOTICE file with any redistribution. The full text is at <https://www.apache.org/licenses/LICENSE-2.0>.

The writing: PolyForm Noncommercial 1.0.0

The chapters of this book are under PolyForm Noncommercial 1.0.0.

- SPDX identifier: PolyForm-Noncommercial-1.0.0
- Released 9 July 2019
- Canonical URL: <https://polyformproject.org/licenses/noncommercial/1.0.0>

Write that URL with no trailing slash. The trailing-slash form returns a 404. It is an ordinary mistake to make and the link then looks broken to whoever you sent it to.

Two things to be straight about:

It is source-available, not open source. PolyForm Noncommercial is **not OSI-approved and not FSF Free/Libre**. Calling it open source would be inaccurate, and this book will not do that.

Licence scanners and corporate allowlists often flag it. If you work somewhere with an automated compliance check, expect this licence to be questioned. I would rather tell you here than let you find it in a review.

The practical effect: you may read this book, quote it, learn from it and use it inside a non-commercial project. You may not sell it or fold it into something you sell. The design system next to it has no such restriction, which is the whole point of splitting them.

The typefaces: SIL OFL 1.1

Four font files ship with this system. Three are subsets — copies with the unused glyphs stripped out to cut the file size — and one is a whole face: the desktop file, which has to carry every glyph because it is meant to be installed and used for anything.

The table below is generated from what is actually on disk, so it cannot omit a file. It used to say "three faces, each as a subset" while a fourth sat in the same directory: the whole IBM Plex Mono, renamed, and the largest single redistributed font in the tree. The licence obligations were met for it — the OFL text sits beside it and the rename is what clause 3 requires — but it was named on no licence surface, and it is the file a reader is most likely to pass on to somebody else.

Every font artefact this kit redistributes: 3 subsets and 1 whole face, each with its licence and its size as written to disk. The whole face is the desktop file, listed here because a reader is more likely to install or pass on that one than any subset.

Family	Licence	Reserved Font Name	Form	Size on disk	Licence text
Literata	SIL OFL 1.1	No	Subset	82 kB	fonts/literata-OFL.txt
Noto Serif Bengali	SIL OFL 1.1	No	Subset	111 kB	fonts/notoserifbengali-OFL.txt
Aninda Mono	SIL OFL 1.1	Yes — renamed	Subset	10 kB	fonts/anindamono-OFL.txt
Aninda Mono (desktop)	SIL OFL 1.1	Yes — renamed	Whole face	136 kB	fonts/anindamono-OFL.txt

The Open Font Licence is at version **1.1, dated 26 February 2007**. **There is no version 1.2**. The canonical home is now `openfontlicense.org`, moved from `scripts.sil.org`.

The Reserved Font Name, and why one face was renamed

A font under the OFL may carry a **Reserved Font Name** — a name that a modified copy is not allowed to keep. Subsetting a font counts as modification under clause 3 of the licence, so a subset of a font with a Reserved Font Name has to be renamed before it is distributed.

IBM Plex Mono carries the Reserved Font Name "Plex". The subset used in this system is therefore renamed **Aninda Mono** in its name table, and the build refuses to finish if any Reserved Font Name survives the rename.

Literata and Noto Serif Bengali carry no Reserved Font Name, so their subsets keep their real names. Renaming those would make the system harder to trace rather than safer, which is the opposite of what the rule is for.

Each subset ships with the full OFL .txt of the font it came from.

The identity: not licensed at all

The name "Aninda Studio", the Bangla name, the mark, the wordmark, the tile and any lockup of them are not licensed. No permission to use them is granted by this book, by the Apache licence on the system, or by the PolyForm licence on the writing.

They are not included in the npm or Python packages, and no licence to them is granted by those packages either.

Fork the system and put your own identity on it. That is the intended path and it is a supported one: the design system was built so that it works with the mark removed. If deleting the identity broke the tokens, the boundary would have been drawn in the wrong place.

What you may do without asking

- Use the design system, in full, in anything.
- Reproduce the mark to refer to Aninda Studio — in a piece of writing about the studio, in a list of tools you use, in a credit line.
- Link to the studio.

What needs a written yes from me

- Putting the mark on a product, a package, a site or a document that is not Aninda Studio's.
- Any use that suggests I made, endorsed or checked something I did not.
- Modifying the mark in any way, including recolouring it outside the rules in chapter 3.

Write to aninda.sh15@gmail.com. I answer.

The name is an adjective

Never a verb, never a plural, never a possessive. *An Aninda Studio component* — yes. *Anindas* — no. That convention is borrowed from Apple's own trademark guidance, and it is the ordinary way a name stays a name.

Contact

aninda.sh15@gmail.com

Not legal advice.

Bangla · বাংলা

The title of this chapter is verified. Its body prose is not written.

Only Bangla that has been checked against the Bangla Academy's 2012 spelling rules and its own dictionary is shipped in this system. No Bangla is written for this book. Where a verified string does not exist, the English stays and the gap is named here and in chapter 14.

The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

A licence chapter is the last place to put unreviewed wording. A licence term that is slightly off in one language and exact in the other is worse than a licence term that appears in one language only, because the reader cannot tell which of the two binds.

The English section is the binding text. Both names below are proper names and are written in Latin in every language.

Each part of the kit and its licence.	
Part	Terms
The design system	Apache-2.0
The writing	PolyForm-Noncommercial-1.0.0
The typefaces	OFL-1.1

অনিন্দ্য স্টুডিও — the name, the mark, the wordmark, the tile, the lockups Not licensed

Contact: aninda.sh15@gmail.com

Not legal advice.

What this system does not do • যা এই পদ্ধতি করে না

The honest list. Not a disclaimer — every item changes how much weight to put on something earlier.

English

This chapter is not a disclaimer. Every item below is a real gap that changes how much weight you should put on something earlier in the book.

No screen-reader testing by a person who depends on one

Contrast is computed. Target sizes are measured. Focus rings are read off differenced pixel buffers. Roles and labels are in the markup and were reviewed.

None of that is the same claim as "this works with a screen reader". Nobody who uses a screen reader every day has tried this system. Lived accessibility is a different claim, and this book does not make it. If you depend on one and something here fails you, write to me and I will fix it.

No user research

Not a small study. None.

Google's Material 3 Expressive is described on Google's own site as backed by 46 studies and more than 18,000 participants. That figure is reported here as **Google's stated research**: there is no published methodology, no preregistration, no peer review, and the page carrying the claim has no publish date.

This kit has no equivalent, and no amount of rigour elsewhere substitutes for it. Every judgement in this book about how something *feels*— whether the corrected Bangla size looks equal to the Latin beside it, whether 220 ms feels right, whether the mark reads as a river — is one person's judgement.

The Bangla has not been reviewed by a second Bangla reader

This is the single most important gap in the book.

The Bangla was checked rule by rule against the Bangla Academy's 2012 spelling rules and its own dictionary, with page numbers recorded for every ruling. That is a careful reading of primary sources by one person. It is not a review by a second Bangla reader, and it cannot substitute for one.

The type work has the same shape. HarfBuzz shaping produced no dotted circles, no missing glyphs and no stray hasantas across sixteen conjuncts in all ten Bangla faces, and pixel analysis confirmed the

মাত্রা is continuous. **That proves nothing is broken. It does not prove the Bangla is good.** Whether Noto Serif Bengali's conjuncts read as well-drawn Bangla to someone in Barishal is a judgement no measurement can make.

Where no verified string existed, the English was left in place and the gap was named rather than filled. Those gaps are listed with the chapter they fall in, below.

Chromium only

Every rendered measurement in this system — the type metrics, the মাত্রা continuity, the line-height collision floors, the contrast readings, the focus ring — comes from **headless Chromium**, at a device scale factor of 1 for measurement and 2 for the specimen images. It runs on macOS here and on Ubuntu in CI, from a clean checkout, so the readings are not particular to one machine.

Windows uses DirectWrite. Android has its own stack. Hinting and stem darkening differ between them. The findings most likely to move are the sub-pixel মাত্রা results at 11 px and 12 px. Safari and Firefox were not tested at all.

Nothing was tested in print, either. The device-pixel argument that sets the 12 px Bangla floor does not apply on paper, so print can go smaller than this system says.

The icon decision's untested dynamic cost

One rounded icon is used on every surface, Apple included. One part of the static cost is measured: under the circle watchOS and visionOS mask to, the rounded icon and the square master are the same image in every pixel, so there the penalty is nil. The other part is a judgement, not a measurement — under Apple's rounded-rectangle mask the difference looks slight to me in a static render, and it cannot be measured, because Apple publishes no corner radius and so there is no mask to composite against without substituting this kit's own radius for Apple's and measuring the substitution. Chapter 04 separates the two at length, and 04_mark/manifest.json records the same split.

The dynamic cost was not measured and is not known. Liquid Glass specular highlights are generated at run time by Apple's own renderer from the layer edges, and there is no way to reproduce that outside it. The rounded artwork may produce a highlight that follows the wrong geometry on a moving device. It might be invisible. Saying which would be a guess, and this book will not guess.

Smaller things, named rather than buried

- **Conjunct coverage is a sample.** Sixteen conjuncts and five words were tested, not the full combinatorial set of Bengali conjuncts.
- **No fallback or email testing.** How these type stacks degrade when the webfonts fail to load was not tested, and email clients cannot use webfonts at all.
- **Licence reading is not legal advice.** Every OFL.txt was read in full and the Reserved Font Name declarations were extracted mechanically. I am confident in the reading. It is still a reading.

- **Guidance moves.** Apple's icon guidance changed materially twice in two years. Two benchmark systems were archived in the ten weeks before this kit's research pass, and a Figma deprecation landed three days after it.
- **The PDF has no bookmark tree.** The print pipeline is Playwright only, with no post-processing tool, and there is no way to add PDF bookmarks afterwards. A generated table of contents with internal links stands in for it.
- **This system has one user.** It has never been handed to a second designer to build with. Everything about how learnable it is is untested.

The Bangla gaps in this book, by chapter

All 14 chapters have a verified Bangla title. All 14 lack verified Bangla body prose — only the approved strings appear.

Chapter	Verified Bangla title	What is missing
01 — Welcome	স্বাগতম	Body prose. Only the verified strings appear.
02 — The name	নাম	Body prose. Only the verified strings appear.
03 — The mark	চিহ্ন	Body prose. Only the verified strings appear.
04 — Icons	আইকন	Body prose. Only the verified strings appear.
05 — Colour	রং	Body prose. Only the verified strings appear.
06 — Type	হরফ	Body prose. Only the verified strings appear.
07 — Space and shape	ফাঁক ও আকার	Body prose. Only the verified strings appear.
08 — Components	উপাদান	Body prose. Only the verified strings appear.
09 — Motion	গতি	Body prose. Only the verified strings appear.
10 — Voice	কণ্ঠস্বর	Body prose. Only the verified strings appear.
11 — Writing in Bangla	বাংলায় লেখা	Body prose. Only the verified strings appear.
12 — Applying it	কাজে লাগানো	Body prose. Only the verified strings appear.
13 — Licence and trademarks	লাইসেন্স ও ট্রেডমার্ক	Body prose. Only the verified strings appear.
14 — What this system does not do	যা এই পদ্ধতি করে না	Body prose. Only the verified strings appear.

The two composed theme labels, with the basis each rests on read from 06_type/bangla-strings.json. Both are approved; an earlier version of this book said they were not, over a form nobody had reviewed.

Label	String used	Standing
High contrast, light	বেশি কনট্রাস্ট, আলো	Verified th-3 (বেশি কনট্রাস্ট) joined to verified th-1 (আলো) with the same comma the English label uses. Comma is ordinary Western punctuation in Bangla — W3C Bengali orthography notes.
High contrast, dark	বেশি কনট্রাস্ট, অন্ধকার	Verified th-3 joined to verified th-2, same construction as theme.hc-light.

All 30 component cards carry a verified Bangla name, so there is no gap to list here.

All 30 component cards carry a verified Bangla subtitle, so there is no gap to list here.

What would close these

In the order I would do them, given the chance:

1. A Bangla reader who is not me, going through every string and every specimen.
2. A person who uses a screen reader daily, going through the components.
3. Rendering checks on Windows and on Android, at 1× and 2×.
4. Five people who are not me, building something small with the kit.

Until those happen, the honest description of this system is: carefully measured, thoroughly documented, and checked by one person — on two operating systems, but in one browser engine, and by nobody who depends on a screen reader.

Where every outside claim comes from

This book leans on outside authorities in a few dozen places — Apple's minimum text sizes, Android's touch target, WCAG's contrast floors, the Bangla Academy's spelling rules, the DTCG format. Until 19 August 2026 it carried exactly two URLs, both of them licence texts, and none of those claims cited anything.

That was the worst omission in the book, because it is the one that asks a reader to take a number on trust while the rest of the system is built on refusing to do that. The table below is the whole apparatus, generated from the same record the benchmark uses, and the benchmark itself now travels inside this file.

Every claim in this book about an outside body rests on one of the 64 sources below. Each carries the URL it was read at and the date the source itself showed, which is not the same as the date it was read: a page with no change log is marked as current at the check rather than given a false date.

All checked 14 August 2026, re-checked 26 August 2026. The full benchmark that produced them, with what each source was read for, is 01_research/BENCHMARK.md, which travels inside this file.

Apple

21 sources from Apple.

Source	URL	Date on the source
Human Interface Guidelines — App Icons	developer.apple.com/design/human-interface-guidelines/app-icons	Change log 8 Jun 2026 (prev. 9 Jun 2025, 10 Jun 2024)
Human Interface Guidelines — Materials	developer.apple.com/design/human-interface-guidelines/foundations/materials/	Current at check
Human Interface Guidelines — Typography	developer.apple.com/design/human-interface-guidelines/typography	Change log 16 Dec 2025
Human Interface Guidelines — Accessibility	developer.apple.com/design/human-interface-guidelines/accessibility	Current at check
Human Interface Guidelines — Motion	developer.apple.com/design/human-interface-guidelines/motion	Current at check
Human Interface Guidelines — Focus and selection	developer.apple.com/design/human-interface-guidelines/focus-and-selection	Change log 24 Oct 2023
Human Interface Guidelines — Branding	developer.apple.com/design/human-interface-guidelines/foundations/branding/	Current at check
Adopting Liquid Glass	developer.apple.com/documentation/technologyoverviews/adopting-liquid-glass	Current at check
Creating your app icon using Icon Composer (Xcode documentation)	<a href="https://developer.apple.com/documentation/xcode`-Icon-Composer-article">developer.apple.com/documentation/xcode` – Icon Composer article	Current at check
Apple Design Resources (App Icon Template)	developer.apple.com/design/resources/	Refreshed 23 Jun 2026
App Store Marketing Guidelines	developer.apple.com/app-store/marketing/guidelines/	Current at check
Apple Trademark List	apple.com/legal/intellectual-property/trademark/appletmlist.html	States updates as of 14 Jul 2026
Guidelines for Using Apple Trademarks and Copyrights	apple.com/legal/intellectual-property/guidelinesfor3rdparties.html	Current at check
Apple Identity Guidelines for Channel Affiliates (PDF)	apple.com/legal/sales-support/certification/docs/logo_guidelines.pdf	Content c. 2013
Apple Style Guide (PDF)	help.apple.com/pdf/applestyleguide/en_US/apple-style-guide.pdf	Re-published 25 Jun 2026
WWDC26 session 251 — *Communicate your brand identity on iOS*	developer.apple.com/videos/play/wwdc2026/251/	8 Jun 2026
App Store asset best practices	developer.apple.com/app-store/asset-best-practices/	Announced 5 Aug 2026; read 26 Aug 2026

Source	URL	Date on the source
App Store Connect — screenshot specifications	developer.apple.com/help/app-store-connect/reference/app-information/screenshot-specifications	Read 26 Aug 2026
App Store Connect — app store localizations	developer.apple.com/help/app-store-connect/reference/app-store-localizations	Bangla added 30 Mar 2026; read 26 Aug 2026
Accessibility Nutrition Labels — overview	developer.apple.com/help/app-store-connect/manage-app-accessibility/overview-of-accessibility-nutrition-labels	Read 26 Aug 2026
What's new in Apple design guidance	developer.apple.com/design/whats-new/	Newest entry 23 Jun 2026; read 26 Aug 2026

18 sources from Google.

Source	URL	Date on the source
Material 3 Expressive launch announcement	blog.google/products-and-platforms/platforms/android/material-3-expressive-android-wearos-launch/	13 May 2025
Material 3 specification	m3.material.io	Current at check
Compose Material 3 release notes (versions, alphas, promotions)	developer.android.com/jetpack/androidx/releases/compose-material3	1.4.0 and 1.5.0-alpha26, both 12 Aug 2026
Compose `ColorScheme`, `ShapeTokens.kt`, `MaterialShapes.kt`, `MotionScheme`	AndroidX source, via `developer.android.com` and the AndroidX repository	Checked 14 Aug 2026
Material Components for Android (maintenance mode)	github.com/material-components/material-components-android	Checked 14 Aug 2026
Material Web Components (maintenance mode)	github.com/material-components/material-web	Checked 14 Aug 2026
`material-color-utilities`	github.com/material-foundation/material-color-utilities	Last push 10 Aug 2026
Material Theme Builder (repository archived)	github.com/material-foundation/material-theme-builder	Archived Jul 2026
Android accessibility guidance (touch targets, contrast)	developer.android.com – accessibility pages	Checked 14 Aug 2026
Expressive design research	design.google/library/expressive-material-design-google-research	**No publish date on the page**
Google brand resources (redirect chain)	google.com/permissions → about.google/brand-resource-center → partnermarketinghub.withgoogle.com ; brand.google → sign-in	Checked 14 Aug 2026
Google Fonts licensing structure	fonts.google.com and github.com/google/fonts (`ofl/`, `apache/`, `ufl/`)	Checked 14 Aug 2026
Google Play — icon design specifications	developer.android.com/distribute/google-play/resources/icon-design-specifications	Page updated 15 Jun 2026; read 26 Aug 2026
Android — adaptive icon design	developer.android.com/develop/ui/views/launch/icon_design_adaptive	Page updated 13 Aug 2026; read 26 Aug 2026
Google Play Console — add preview assets	support.google.com/googleplay/android-developer/answer/9866151	No date published; read 26 Aug 2026
Android — core app quality guidelines	developer.android.com/docs/quality-guidelines/core-app-quality	Page updated 21 Aug 2026; read 26 Aug 2026
Material 3 — colour roles	m3.material.io/styles/color/roles	Read 26 Aug 2026 through a browser; the page is JavaScript-rendered
androidx — StandardMotionTokens.kt and ExpressiveMotionTokens.kt	github.com/androidx/androidx	VERSION v0.14.0; read 26 Aug 2026

Standards and formats

8 sources from Standards and formats.

Source	URL	Date on the source
WCAG 2.2	w3.org/TR/WCAG22/	W3C Recommendation, 12 Dec 2024
WCAG 3.0	w3.org/TR/wcag-3.0/	Working Draft, 3 Mar 2026
Design Tokens Format Module 2025.10	designtokens.org/tr/2025.10/format/	Final Community Group Report, 28 Oct 2025
DTCG schema identifier used by implementers	tr.designtokens.org/format/	Checked 14 Aug 2026
SIL Open Font Licence 1.1	openfontlicense.org	Licence 26 Feb 2007; FAQ 1.1-update7, Nov 2023
Apache License 2.0	apache.org/licenses/LICENSE-2.0	Checked 14 Aug 2026
PolyForm Noncommercial 1.0.0	polyformproject.org/licenses/noncommercial/1.0.0` (**no trailing slash**)	Released 9 Jul 2019
SPDX licence list	spdx.org/licenses/	Checked 14 Aug 2026

Tooling and other design systems

17 sources from Tooling and other design systems.

Source	URL	Date on the source
IBM Carbon `@carbon/themes` (DTCG tokens)	github.com/carbon-design-system/carbon	11.79.0, 12 Aug 2026, Apache-2.0
Style Dictionary	styledictionary.com	5.5.1, 7 Aug 2026, Apache-2.0
Tokens Studio for Figma	Tokens Studio documentation	Checked 14 Aug 2026, MIT
Figma REST and Plugin APIs; Variables API plan requirements	developers.figma.com/docs/	Checked 14 Aug 2026
Figma Code Connect — parser deprecation	developers.figma.com/docs/code-connect/` and `github.com/figma/code-connect	Framework-specific parsers unsupported from 17 Aug 2026
GitHub Primer	primer.style	Checked 14 Aug 2026, MIT
Atlassian `@atlassian/tokens`	atlassian.design	16.7.0, Apache-2.0
Adobe Spectrum design data (repository renamed)	github.com/adobe/spectrum-design-data	Checked 14 Aug 2026, Apache-2.0
Microsoft Fluent `@fluentui/tokens`	github.com/microsoft/fluentui	1.0.0-alpha.24
Shopify Polaris	polaris.shopify.com	Polaris React archived and deprecated 11 Aug 2026; Polaris Web Components released 1 Oct 2025
Salesforce Lightning Design System	github.com/salesforce-ux/design-system	Repository archived, last push 2 Jun 2026; npm 2.264.0, 22 Jul 2026, BSD-3-Clause
GOV.UK Design System and brand site	design-system.service.gov.uk` and `brand.design-system.service.gov.uk	Code MIT; content OGL v3.0; assets "All rights reserved"
Uber Base	base.uber.com	Checked 14 Aug 2026, MIT
Meta Brand Resource Center	meta.com/brand/resources/	Checked 14 Aug 2026
Mailchimp content style guide	styleguide.mailchimp.com	Repository last pushed May 2022; no LICENSE file
Stripe	No Stripe-owned brand or design-system page reachable	Checked 14 Aug 2026
Spotify Encore	No public documentation found	Checked 14 Aug 2026

Bangla · বাংলা

The title of this chapter is verified. Its body prose is not written.

Only Bangla that has been checked against the Bangla Academy's 2012 spelling rules and its own dictionary is shipped in this system. No Bangla is written for this book. Where a verified string does not exist, the English stays and the gap is named here and in chapter 14.

The English section of this chapter carries the full text. Whatever Bangla appears below is verified, and there is no more of it.

This chapter's own subject is the reason it is short. The largest single gap in the system is that **no second Bangla reader has reviewed any of the Bangla**. Writing a full Bangla chapter here, unreviewed, would be the exact failure the chapter exists to name.

The verified title of this chapter is যা এই পদ্ধতি করে না, and it is one of the six strings the Bangla review changed.

Every Bangla gap in this book

All 14 chapters have a verified Bangla title. All 14 lack verified Bangla body prose — only the approved strings appear.

Chapter	Verified Bangla title	What is missing
01 — Welcome	স্বাগতম	Body prose. Only the verified strings appear.
02 — The name	নাম	Body prose. Only the verified strings appear.
03 — The mark	চিহ্ন	Body prose. Only the verified strings appear.
04 — Icons	আইকন	Body prose. Only the verified strings appear.
05 — Colour	রং	Body prose. Only the verified strings appear.
06 — Type	হরফ	Body prose. Only the verified strings appear.
07 — Space and shape	ফাঁক ও আকার	Body prose. Only the verified strings appear.
08 — Components	উপাদান	Body prose. Only the verified strings appear.
09 — Motion	গতি	Body prose. Only the verified strings appear.
10 — Voice	কণ্ঠস্বর	Body prose. Only the verified strings appear.
11 — Writing in Bangla	বাংলায় লেখা	Body prose. Only the verified strings appear.
12 — Applying it	কাজে লাগানো	Body prose. Only the verified strings appear.
13 — Licence and trademarks	লাইসেন্স ও ট্রেডমার্ক	Body prose. Only the verified strings appear.
14 — What this system does not do	যা এই পদ্ধতি করে না	Body prose. Only the verified strings appear.

The two composed theme labels, with the basis each rests on read from 06_type/bangla-strings.json. Both are approved; an earlier version of this book said they were not, over a form nobody had reviewed.

Label	String used	Standing
High contrast, light	বেশি কনট্রাস্ট, আলো	Verified th-3 (বেশি কনট্রাস্ট) joined to verified th-1 (আলো) with the same comma the English label uses. Comma is ordinary Western punctuation in Bangla — W3C Bengali orthography notes.
High contrast, dark	বেশি কনট্রাস্ট, অন্ধকার	Verified th-3 joined to verified th-2, same construction as theme.hc-light.

All 30 component cards carry a verified Bangla name, so there is no gap to list here.

All 30 component cards carry a verified Bangla subtitle, so there is no gap to list here.

Every file, in this file

74 files, 11.0 MB on disk.

The download links are stripped in the print build. Their names are kept. A paper page cannot be clicked, so the payloads buy a printed reader nothing while making the PDF about ten times larger. Chapter 12 gives the measured figures. Open `Aninda-Studio-Guidebook.html` for the files themselves.

- `07_tokens/build/primitive.tokens.json` — Design tokens, DTCG 2025.10 · 42 kB
- `07_tokens/build/semantic.light.tokens.json` — Design tokens, DTCG 2025.10 · 15 kB
- `07_tokens/build/semantic.dark.tokens.json` — Design tokens, DTCG 2025.10 · 15 kB
- `07_tokens/build/semantic.hc-light.tokens.json` — Design tokens, DTCG 2025.10 · 16 kB
- `07_tokens/build/semantic.hc-dark.tokens.json` — Design tokens, DTCG 2025.10 · 16 kB
- `07_tokens/build/forced-colors.map.json` — Which system colour each role gives way to in forced colours · 2 kB
- `07_tokens/css/tokens.css` — Every token as a CSS custom property, in five layers · 9 kB
- `01_research/BENCHMARK.md` — The benchmark this book cites: every external source with its URL and date, and the 28 acceptance criteria with their verdicts · 76 kB
- `08_components/src/components.css` — The component layer. No literal colour in it · 34 kB
- `08_components/fonts/literata-subset.woff2` — Literata subset, SIL OFL 1.1 · 82 kB
- `08_components/fonts/literata-OFL.txt` — The full licence text for Literata · 4 kB
- `08_components/fonts/notoserifbengali-subset.woff2` — Noto Serif Bengali subset, SIL OFL 1.1 · 111 kB
- `08_components/fonts/notoserifbengali-OFL.txt` — The full licence text for Noto Serif Bengali · 4 kB
- `08_components/fonts/anindamono-subset.woff2` — Aninda Mono subset, SIL OFL 1.1 · 10 kB
- `08_components/fonts/anindamono-OFL.txt` — The full licence text for Aninda Mono · 4 kB
- `08_components/fonts/AnindaMono-Regular.ttf` — Aninda Mono (desktop) whole face, not subset, SIL OFL 1.1 · 136 kB
- `04_mark/svg/mark-regular.svg` — Identity artwork · 457 bytes
- `04_mark/svg/mark-heavy.svg` — Identity artwork · 457 bytes
- `04_mark/svg/wordmark-latin.svg` — Identity artwork · 25 kB
- `04_mark/svg/wordmark-bangla.svg` — Identity artwork · 28 kB
- `04_mark/svg/tile-web.svg` — Identity artwork · 495 bytes
- `04_mark/svg/icon-1024.svg` — Identity artwork · 527 bytes
- `04_mark/svg/icon-512.svg` — Identity artwork · 519 bytes
- `04_mark/svg/icon-192.svg` — Identity artwork · 507 bytes

- [04_mark/svg/icon-apple-1024.svg](#) — Identity artwork · 545 bytes
- [04_mark/svg/icon-apple-1088-watch.svg](#) — Identity artwork · 512 bytes
- [04_mark/svg/icon-apple-1024-dark.svg](#) — Identity artwork · 509 bytes
- [04_mark/svg/icon-apple-1024-mono.svg](#) — Identity artwork · 478 bytes
- [04_mark/svg/icon-android-background-108.svg](#) — Identity artwork · 258 bytes
- [04_mark/svg/icon-android-foreground-108.svg](#) — Identity artwork · 461 bytes
- [04_mark/svg/icon-android-monochrome-108.svg](#) — Identity artwork · 487 bytes
- [04_mark/manifest.json](#) — The mark's construction, with every check it passed · 10 kB
- [05_colour/generated/estuary.proof.json](#) — Every colour, every ramp, every measured ratio · 43 kB
- [12_packages/npm/dist/index.cjs](#) — Published package build output · 245 bytes
- [12_packages/npm/dist/index.d.ts](#) — Published package build output · 251 bytes
- [12_packages/npm/dist/index.mjs](#) — Published package build output · 264 bytes
- [12_packages/npm/dist/index.tokens.json](#) — Published package build output · 1 kB
- [12_packages/npm/dist/layout.css](#) — Published package build output · 147 bytes
- [12_packages/npm/dist/tokens.css](#) — Published package build output · 9 kB
- [12_packages/npm/dist/tokens.dark.css](#) — Published package build output · 810 bytes
- [12_packages/npm/dist/tokens.hc-dark.css](#) — Published package build output · 816 bytes
- [12_packages/npm/dist/tokens.hc-light.css](#) — Published package build output · 819 bytes
- [12_packages/npm/dist/tokens.light.css](#) — Published package build output · 866 bytes
- [12_packages/npm/dist/typography.css](#) — Published package build output · 403 bytes
- [08_components/cards/foundations/colour.html](#) — Foundations card — Colour · 373 kB
- [08_components/cards/foundations/typography.html](#) — Foundations card — Typography · 345 kB
- [08_components/cards/foundations/space-and-shape.html](#) — Foundations card — Space and shape · 344 kB
- [08_components/cards/foundations/motion.html](#) — Foundations card — Motion · 341 kB
- [08_components/cards/foundations/the-marks.html](#) — Foundations card — The marks · 357 kB
- [08_components/cards/foundations/accessibility.html](#) — Foundations card — Accessibility · 346 kB
- [08_components/cards/components/button.html](#) — Components card — Button · 336 kB
- [08_components/cards/components/input.html](#) — Components card — Input · 337 kB
- [08_components/cards/components/select.html](#) — Components card — Select · 339 kB
- [08_components/cards/components/checkbox-radio.html](#) — Components card — Checkbox and radio · 339 kB
- [08_components/cards/components/textarea.html](#) — Components card — Textarea · 333 kB
- [08_components/cards/components/badge.html](#) — Components card — Badge · 347 kB
- [08_components/cards/components/card.html](#) — Components card — Card · 334 kB
- [08_components/cards/components/alert.html](#) — Components card — Alert · 341 kB

- [08_components/cards/components/dialog.html](#) — Components card — Dialog · 331 kB
- [08_components/cards/components/table.html](#) — Components card — Table · 340 kB
- [08_components/cards/components/tabs.html](#) — Components card — Tabs · 336 kB
- [08_components/cards/components/nav.html](#) — Components card — Nav · 340 kB
- [08_components/cards/components/breadcrumb.html](#) — Components card — Breadcrumb · 331 kB
- [08_components/cards/components/toast.html](#) — Components card — Toast · 341 kB
- [08_components/cards/components/empty-state.html](#) — Components card — Empty state · 336 kB
- [08_components/cards/components/code-block.html](#) — Components card — Code block · 335 kB
- [08_components/cards/patterns/sign-in.html](#) — Patterns card — Sign in · 336 kB
- [08_components/cards/patterns/settings.html](#) — Patterns card — Settings · 344 kB
- [08_components/cards/patterns/dashboard.html](#) — Patterns card — Dashboard · 356 kB
- [08_components/cards/patterns/docs-page.html](#) — Patterns card — Docs page · 341 kB
- [08_components/cards/patterns/landing.html](#) — Patterns card — Landing · 344 kB
- [08_components/cards/patterns/pricing.html](#) — Patterns card — Pricing · 349 kB
- [08_components/cards/patterns/not-found.html](#) — Patterns card — Not found · 333 kB
- [08_components/cards/patterns/form-with-validation.html](#) — Patterns card — Form with validation · 345 kB

Generated by `09_guidebook/build.py` from the sources in this repository. Do not hand-edit this file — change the sources and re-run. `build.py --check` proves it has not drifted.

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